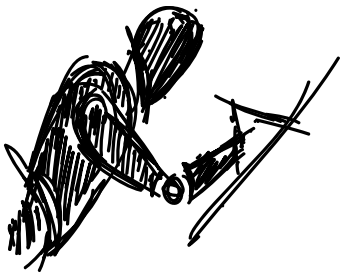

Variational Bayes

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Creative Machines Lab

Final Presentation
Bayesian Scientific Computing, Spring 2013

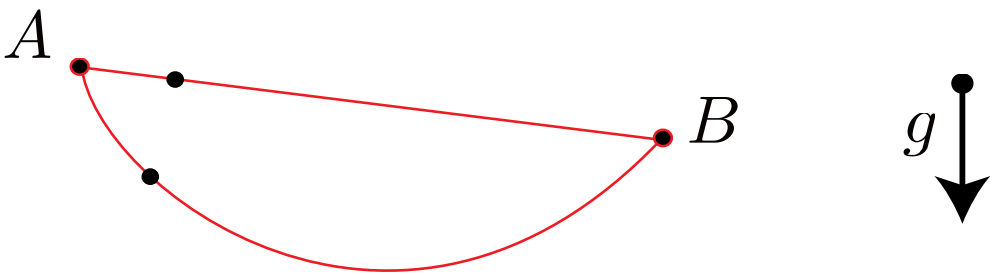
Introduction: What is Variational Bayes?

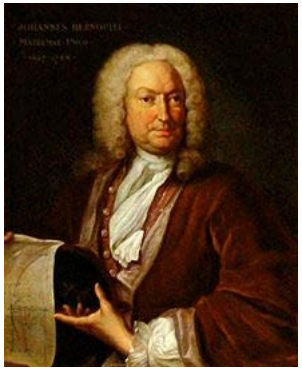
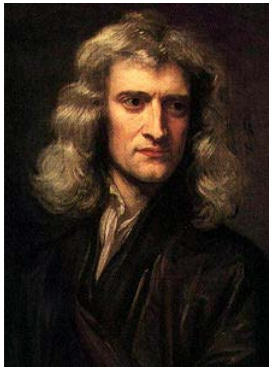
- Optimization applied to the problem of Bayesian Inference
- Foundation in the Calculus of Variations
- A deterministic alternative to MCMC
- Often faster, provides a bound
- Connection to the Expectation Maximization (EM) algorithm

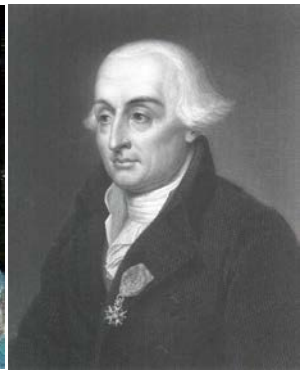
Introduction

What is the *fastest* path between two points?

Brochistrochrone problem





Introduction: Calculus of Variations

Calculus of variations concerns with optimizing w.r.t. functions (not variables)

$$S[q] = \int_b^a L(x, q(x), q'(x))dx$$

We seek a function q(x) that yields an extremum of $S[q]$.

q(x) can be obtained by solving a differential equation satisfying:

$$\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial L}{\partial q'}\right) = 0$$

Equality constraints on q can be introduced using Lagrange multipliers:

One constraint on q may be to require it to be valid probability distribution:

$$\int_b^a \{L(x, q(x))dx\} + \lambda \underbrace{\left(\int_b^a q(x) - 1\right)}_{\sim \text{ subject to } \int_b^a q(x) = 1}$$

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Expectation Maximization Revisited

Expectation Maximization Revisited: E-step

We want to find a maximum likelihood solution for parameters θ

$$P(\mathbf{x}|\theta) = \sum_{\mathbf{z}} P(\mathbf{x}, \mathbf{z}|\theta)$$

$$\log P(\mathbf{x}|\theta) = \log \sum_{\mathbf{z}} q(\mathbf{z}) \frac{P(\mathbf{x}, \mathbf{z}|\theta)}{q(\mathbf{z})}$$

$$\geq \mathbb{E}_q \left[\log \frac{P(\mathbf{x}, \mathbf{z}|\theta)}{q(\mathbf{z})} \right]$$

$$\geq \mathbb{E}_q \left[\log \frac{P(\mathbf{z}|\mathbf{x}, \theta) P(\mathbf{x}|\theta)}{q(\mathbf{z})} \right]$$

$$\geq \mathbb{E}_q \left[\log P(\mathbf{x}|\theta) \right] - \mathbb{E}_q \left[\log \frac{q(\mathbf{z})}{P(\mathbf{z}|\mathbf{x}, \theta)} \right]$$

$$\geq \mathbb{E}_q \left[\log P(\mathbf{x}|\theta) \right] - D_{\mathrm{KL}}(q(\mathbf{z}) \| P(\mathbf{z}|\mathbf{x}, \theta))$$

$$\geq \log P(\mathbf{x}|\theta) - \underbrace{D_{\mathrm{KL}}(q(\mathbf{z}) \| P(\mathbf{z}|\mathbf{x}, \theta))}_{\text{minimize}}$$

Expectation Maximization Revisited: E-step

$$\log P(\mathbf{x}|\theta) \geq \log P(\mathbf{x}|\theta) - \underbrace{D_{\mathrm{KL}}(q(\mathbf{z})\|P(\mathbf{z}|\mathbf{x},\theta))}_{\text{minimize}}$$

$$\Downarrow$$

$$q(\mathbf{z}) = P(\mathbf{z}|\mathbf{x},\theta)$$

Expectation Maximization Revisited: M-step

$$\log P(\mathbf{x}|\theta) \geq \mathbb{E}_q \left[\log \frac{P(\mathbf{x}, \mathbf{z}|\theta)}{q(\mathbf{z})} \right]$$

$$\log P(\mathbf{x}|\theta) \geq \mathbb{E}_q \left[\log P(\mathbf{x}, \mathbf{z}|\theta) \right] - \mathbb{E}_q \left[\log q(\mathbf{z}) \right]$$

$$\log P(\mathbf{x}|\theta) \geq \mathbb{E}_q \left[\log P(\mathbf{x}, \mathbf{z}|\theta) \right] - H \left[q(\mathbf{z}) \right]$$

$$\Downarrow$$

$$\theta^* = \text{argmax}_\theta \mathbb{E}_q \left[\log P(\mathbf{x}, \mathbf{z}|\theta) \right]$$

$$\text{but } q = P(\mathbf{z}|\mathbf{x}, \theta_{old})$$

$$\theta^* = \text{argmax}_\theta \sum_{\mathbf{z}} P(\mathbf{z}|\mathbf{x}, \theta_{old}) \log P(\mathbf{x}, \mathbf{z}|\theta)$$

Intractable Posterior

What if the posterior is intractable?

approximate it!

$$\Downarrow$$

$$q(\mathbf{z}) = \prod_{i=1}^{M} q_i(\mathbf{z}_i)$$

$$\log P(\mathbf{x}) = \mathcal{L}(q(\mathbf{z})) + D_{\mathrm{KL}}(q(\mathbf{z}) \| P(\mathbf{z}|\mathbf{x}))$$

Find *q(**z**)* that will maximize the <u>lower bound</u> $\mathcal{L}$

(minimize KL divergence)

Intractable Posterior

$$\log P(\mathbf{x}) = \mathcal{L}(q(\mathbf{z})) + D_{\mathrm{KL}}(q(\mathbf{z}) \| P(\mathbf{z}|\mathbf{x}))$$

From before:

$$D_{\mathrm{KL}} = -\sum_{\mathbf{z}} q(\mathbf{z}) \log \frac{P(\mathbf{z}|\mathbf{x})}{q(\mathbf{z})}$$

Then it follows that:

$$\mathcal{L}(q) = \sum_{\mathbf{z}} q(\mathbf{z}) \log \frac{P(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})}$$

Then substitute in our (mean-field) approximation for ***q(z):***

$$q(\mathbf{z}) = \prod_{i=1}^{M} q_i(\mathbf{z}_i)$$

Mean-field Approximation

$$q(\mathbf{z}) = \prod_{i=1}^{M} q_i(\mathbf{z}_i) \quad \Longrightarrow \quad \mathcal{L}(q) = \sum_{\mathbf{z}} q(\mathbf{z}) \log \frac{P(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})}$$

$$\mathcal{L}(q) = \sum_{\mathbf{z}} \prod_{i} q_i \left\{ \log P(\mathbf{x}, \mathbf{z}) - \sum_{i} \log q_i \right\}$$

We will optimize each q_i independently.

$$\mathcal{L}(q) = \sum_{\mathbf{z}_j} q_j \left\{ \sum_{\mathbf{z}_i} \log P(\mathbf{x}, \mathbf{z}) \prod_{i \neq j} q_i \right\} - \sum_{\mathbf{z}_j} q_j \log q_j + \text{const}$$

$$= \sum_{\mathbf{z}_j} q_j \log \tilde{P}(\mathbf{x}, \mathbf{z}_j) - \sum_{\mathbf{z}_j} q_j \log q_j + \text{const}$$

where we define:

$$\log \tilde{P}(\mathbf{x}, \mathbf{z}_j) = \mathbb{E}_{i \neq j} \left[\log P(\mathbf{x}, \mathbf{z}) \right] + \text{const}$$

Mean-field Approximation

$$\mathcal{L}(q) = \sum_{\mathbf{z}_j} q_j \log \tilde{P}(\mathbf{x}, \mathbf{z}_j) - \sum_{\mathbf{z}_j} q_j \log q_j + \text{const}$$

Observe that for q_j maximizing the lower bound implies:

$$\sum_{\mathbf{z}_j} q_j \log \tilde{P}(\mathbf{x}, \mathbf{z}_j) = \sum_{\mathbf{z}_j} q_j \log q_j$$

And since from before:

$$\log \tilde{P}(\mathbf{x}, \mathbf{z}_j) = \mathbb{E}_{i \neq j} \left[\log P(\mathbf{x}, \mathbf{z}) \right] + \text{const}$$

Then the optimal q_j:

$$\boxed{\log q_j^*(\mathbf{z}_j) = \mathbb{E}_{i \neq j} \left[\log P(\mathbf{x}, \mathbf{z}) \right] + \text{const}}$$

Mean-field Approximation

$$\log q_j^*(\mathbf{z}_j) = \mathbb{E}_{i \neq j}\left[\log P(\mathbf{x}, \mathbf{z})\right] + \underbrace{\text{const}}_{-\log(Z)}$$

This optimal *q* came out because of our choice for *KL:*

$$D_{\mathrm{KL}}(q \| p) = \mathbb{E}_q\left[\log\left(\frac{q(\mathbf{z})}{P(\mathbf{z}|\mathbf{x})}\right)\right]$$

$$D_{\mathrm{KL}}(q \| p) = \int q(\mathbf{z}) \log\left(\frac{q(\mathbf{z})}{P(\mathbf{z}|\mathbf{x})}\right) d\mathbf{z}$$

Mean-field Approximation: KL divergence

$$D_{\mathrm{KL}}(q\|p) = \int q(\mathbf{z}) \log\left(\frac{q(\mathbf{z})}{P(\mathbf{z})}\right) d\mathbf{z}$$

But what about …

$$D_{\mathrm{KL}}(p\|q) = \int P(\mathbf{z}) \log\left(\frac{P(\mathbf{z})}{q(\mathbf{z})}\right) d\mathbf{z}$$

Deriving the optimal q_i for the reverse KL is a bit more involved…

⇨ As before, start by plugging in q_i into the new KL ($p//q$):

$$D_{\mathrm{KL}}(p\|q) = \underbrace{\int P(\mathbf{z}) \log P(\mathbf{z}) d\mathbf{z}}_{\text{const.}} - \int P(\mathbf{z}) \underbrace{\left[\sum \log q_i(\mathbf{z}_i)\right]}_{\text{mean-field factors}} d\mathbf{z}$$

Mean-field Approximation: KL divergence

$$D_{\mathrm{KL}}(p\|q) = \int P(\mathbf{z}) \log P(\mathbf{z}) d\mathbf{z} - \int P(\mathbf{z}) \left[\sum \log q_i(\mathbf{z}_i) \right] d\mathbf{z}$$

$$\begin{aligned}
D_{\mathrm{KL}}(p\|q) &= -\int P(\mathbf{z}) \left[\sum \log q_i(\mathbf{z}_i) \right] d\mathbf{z} + \mathrm{const.} \\
&= -\int \left(P(\mathbf{z}) \log q_j(\mathbf{z}_j) + P(\mathbf{z}) \sum_{i \neq j} \log q_j(\mathbf{z}_i) \right) d\mathbf{z} \\
&= -\int P(\mathbf{z}) \log q_j(\mathbf{z}_j) d\mathbf{z} \\
&= -\int \log q_j(\mathbf{z}_j) \left[\int P(\mathbf{z}) \prod_{i \neq j} d\mathbf{z}_i \right] d\mathbf{z}_j \\
&= -\int \log q_j(\mathbf{z}_j) F_j(\mathbf{z}_j) d\mathbf{z}_j \quad \Longleftarrow
\end{aligned}$$

Maximize w.r.t. q_j !

Mean-field Approximation: KL divergence

Maximize:

$$-\int \log q_j(\mathbf{z}_j) F_j(\mathbf{z}_j) d\mathbf{z}_j$$

Subject to:

$$\int q_j(\mathbf{z}_j) d\mathbf{z}_j = 1$$

Solve by adding a Lagrange multiplier:

$$-\int \log q_j(\mathbf{z}_j) F_j(\mathbf{z}_j) d\mathbf{z}_j + \lambda \left(\int q_j(\mathbf{z}_j) d\mathbf{z}_j - 1 \right)$$

where we defined:

$$F_j(\mathbf{z}_j) = \int p(\mathbf{z}) \prod_{i \neq j} d\mathbf{z}_i$$

Mean-field Approximation: KL divergence

Taking a functional derivative with respect to q_j:

$$-\frac{F_j(\mathbf{z}_j)}{q_j(\mathbf{z}_j)} + \lambda = 0 \qquad \Longrightarrow \qquad \lambda q_j(\mathbf{z}_j) = F_j(\mathbf{z}_j)$$

Integrating both sides with respect to z_j:

$$\int \lambda q_j(\mathbf{z}_j) d\mathbf{z}_j = \int F_j(\mathbf{z}_j) d\mathbf{z}_j$$

$$\lambda = \int \left[\int P(\mathbf{z}) \prod_{i \neq j} d\mathbf{z}_i \right] d\mathbf{z}_j = 1$$

Yielding the optimal q_j:

$$q^*(\mathbf{z}_j) = \int P(\mathbf{z}) \prod_{i \neq j} d\mathbf{z}_i = P(\mathbf{z}_j) \Longleftarrow$$

Marginal of z_j !

Mean-field Approximation: KL divergence

Optimal mean-field approximation depends on the direction of KL

$$q||p \rightarrow \quad \log q_j^*(\mathbf{z}_j) = \mathbb{E}_{i \neq j}\left[\log P(\mathbf{x}, \mathbf{z})\right]$$

$$p||q \rightarrow \quad \log q_j(\mathbf{z}_j) = \log P(\mathbf{z}_j)$$

What does this mean for inference?

Toy Example 1

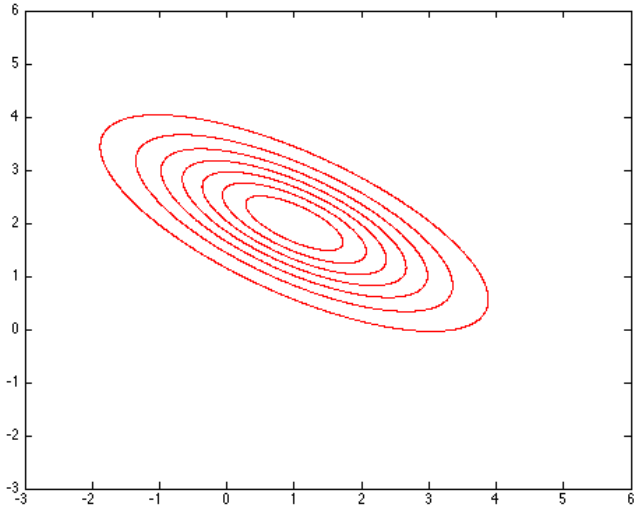
Variational Inference: Toy Example 1 – Approximating a Gaussian

We want to find a mean-field approximation to a 2-D Gaussian:

$$P(\mathbf{z}) = \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1})$$

with $\mathbf{z} = [z_1, z_2]$, and parameters:

$$\boldsymbol{\mu} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \boldsymbol{\Lambda} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$



Variational Inference: Toy Example 1 – Approximating a Gaussian

We want to find a mean-field approximation to a 2-D Gaussian:

$$P(\mathbf{z}) = \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1})$$

We will attempt to find an approximation to $P(\mathbf{z})$ with this factorization:

$$q(z_1, z_2) = q_1(z_1)q_2(z_2)$$

<u>No functional form of the factors is assumed</u>

Using our first definition of KL *q* // *p*, the solution of the optimal factors is:

$$q||p \rightarrow \begin{cases} \log q_1^*(z_1) = \mathbb{E}_{z_2}[\log P(\mathbf{z})] + \text{const.} \\ \log q_2^*(z_2) = \mathbb{E}_{z_1}[\log P(\mathbf{z})] + \text{const.} \end{cases}$$

Variational Inference: Toy Example 1 – Approximating a Gaussian

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad \boldsymbol{\Lambda} = \begin{bmatrix} \Lambda_{11} & \Lambda_{12} \\ \Lambda_{21} & \Lambda_{22} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

Start with the first definition of KL $q||p$, and derive the approximation:

$$q||p \rightarrow \quad \log q^*(z_1) = \mathbb{E}_{z_2}[\log P(\mathbf{z})] + \text{const.}$$

Plug in the original $P(\mathbf{z}) = \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}, \boldsymbol{\Lambda})$ into the expectation above:

$$= \mathbb{E}_{z_2}\left[\log\left(2\pi^2|\boldsymbol{\Lambda}^{-1}|\exp\left(-\frac{1}{2}(\mathbf{z}-\boldsymbol{\mu})^T\boldsymbol{\Lambda}(\mathbf{z}-\boldsymbol{\mu})\right)\right)\right] + \text{const.}$$

$$= \mathbb{E}_{z_2}\left[-\frac{1}{2}\left(\Lambda_{11}(z_1-\mu_1)^2 + 2\Lambda_{12}(z_1-\mu_1)(z_2-\mu_2) + \Lambda_{22}(z_2-\mu_2)^2\right)\right]$$

$$= \mathbb{E}_{z_2}\left[-\frac{1}{2}\left(\Lambda_{11}(z_1^2 - 2\mu_1 z_1 + \mu_1^2) + 2\Lambda_{12}(z_1 z_2 - z_1\mu_2 - z_2\mu_1 + \mu_1\mu_2)\right)\right]$$

$$= -\frac{1}{2}z_1^2\Lambda_{11} + z_1\mu_1\Lambda_{11} - z_1\Lambda_{12}(\mathbb{E}[z_2] - \mu_2) + \text{const.}$$

Variational Inference: Toy Example 1 – Approximating a Gaussian

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad \boldsymbol{\Lambda} = \begin{bmatrix} \Lambda_{11} & \Lambda_{12} \\ \Lambda_{21} & \Lambda_{22} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

Recognize that the expression is quadratic in *z* and is actually a Gaussian:

$$\log q_1^*(z_1) = -\frac{1}{2} z_1^2 \Lambda_{11} + z_1 \mu_1 \Lambda_{11} - z_1 \Lambda_{12} (\mathbb{E}[z_2] - \mu_2) + \text{const.}$$

Complete the square to obtain:

$$q_1^*(z_1) = \mathcal{N}(m_1, \rho_1^{-1})$$

q_1 is a Gaussian with parameters:

$$m_1 = \mu_1 - \frac{\Lambda_{12}}{\Lambda_{11}} (\mathbb{E}[z_2] - \mu_2)$$

$$\rho_1 = \Lambda_{11}$$

Variational Inference: Toy Example 1 – Approximating a Gaussian

By symmetry we obtain the optimal q_2 easily.

$$\Longrightarrow \qquad q_2^*(z_2) = \mathcal{N}(m_2, \rho_1^{-1})$$

$$m_2 = \mu_2 - \frac{\Lambda_{21}}{\Lambda_{22}}(\mathbb{E}[z_1] - \mu_1)$$

$$\rho_2 = \Lambda_{22}$$

q_1 :

$$\Longrightarrow \qquad q_1^*(z_1) = \mathcal{N}(m_1, \rho_1^{-1})$$

$$m_1 = \mu_1 - \frac{\Lambda_{12}}{\Lambda_{11}}(\mathbb{E}[z_2] - \mu_2)$$

$$\rho_1 = \Lambda_{11}$$

<u>Observe that the parameters equations are coupled!</u>

Variational Inference: Toy Example 1 – Approximating a Gaussian

We can perform simple iterative updating:

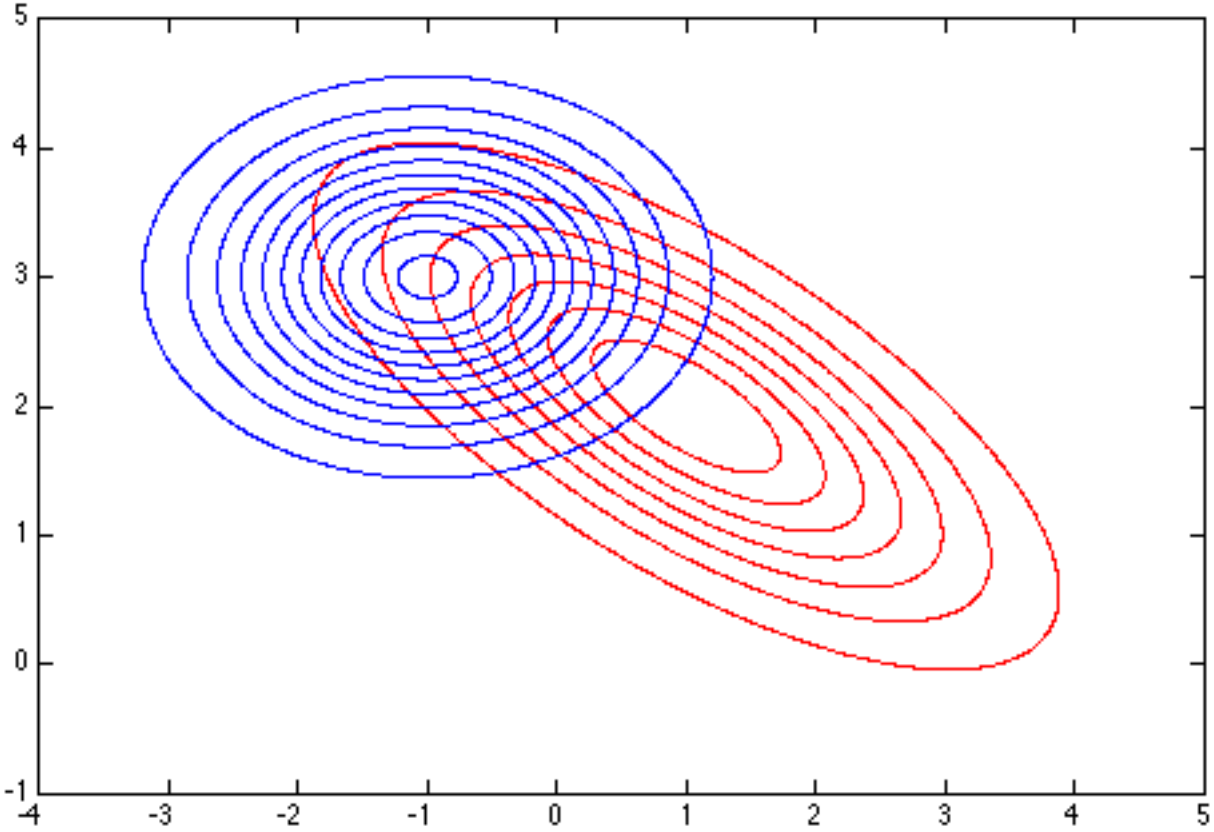
Initialize:

$$\rho_1 \leftarrow \Lambda_{11}$$
$$\rho_2 \leftarrow \Lambda_{22}$$
$$m_1 \leftarrow rand$$
$$m_2 \leftarrow rand$$

While not converged:

$$\left.\begin{aligned} m_1 &\leftarrow \mu_1 - \frac{\Lambda_{12}}{\Lambda_{11}}(\mathbb{E}[z_2] - \mu_2) \\ m_2 &\leftarrow \mu_2 - \frac{\Lambda_{21}}{\Lambda_{22}}(\mathbb{E}[z_1] - \mu_1) \end{aligned}\right\} \text{Linear system}$$

Variational Inference: Toy Example 1 – Approximating a Gaussian



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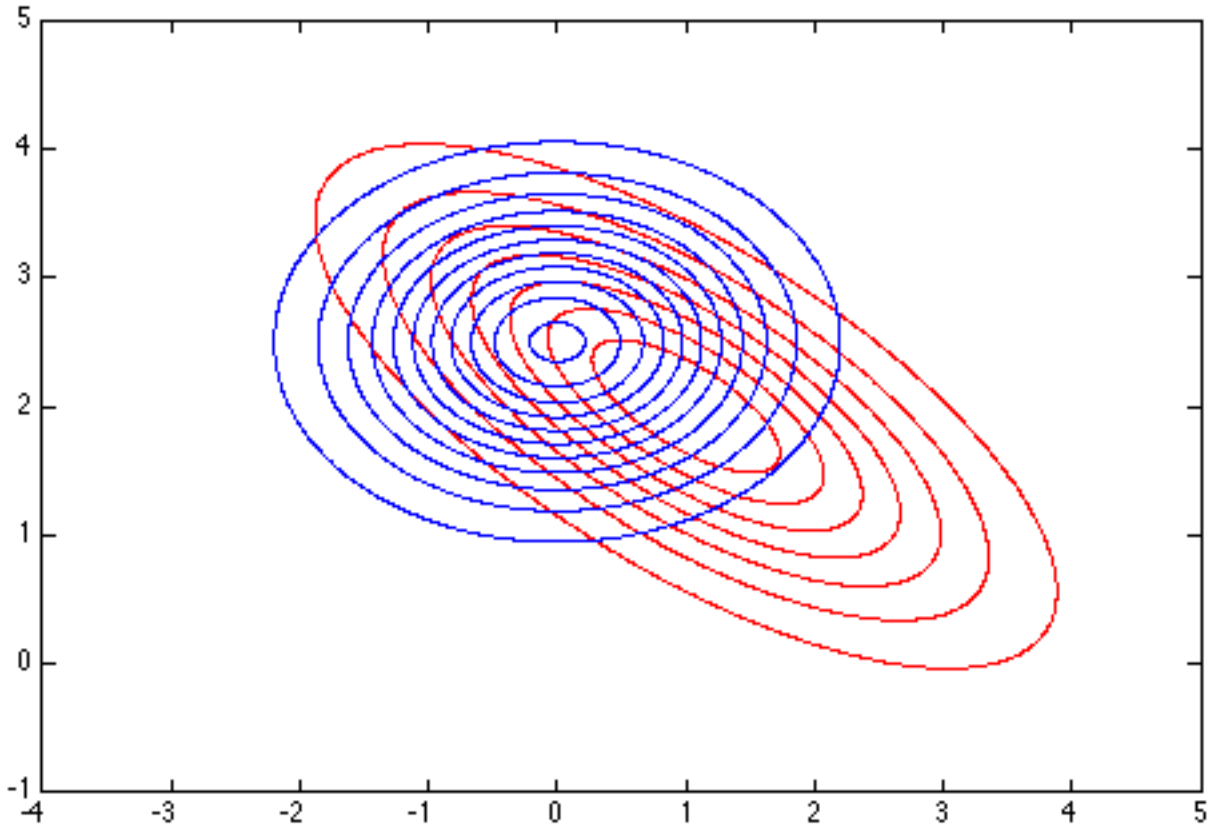

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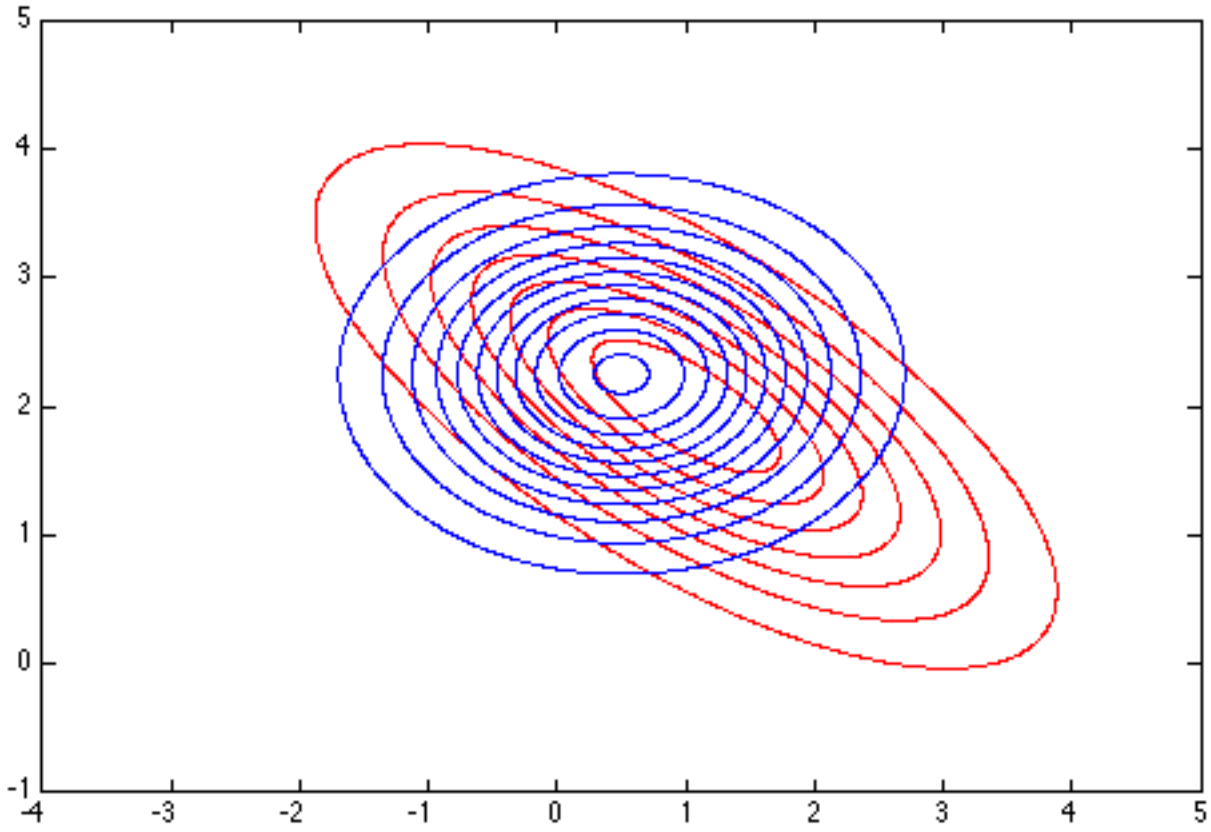

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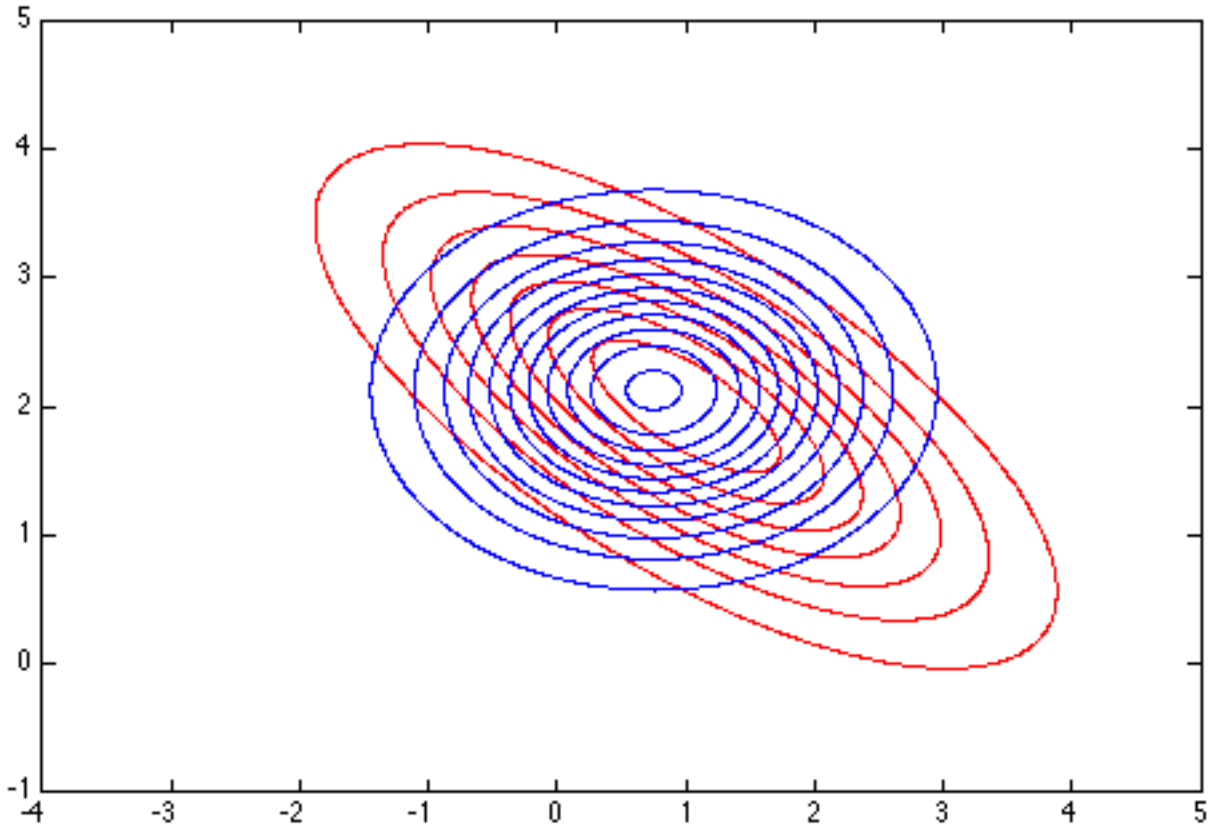

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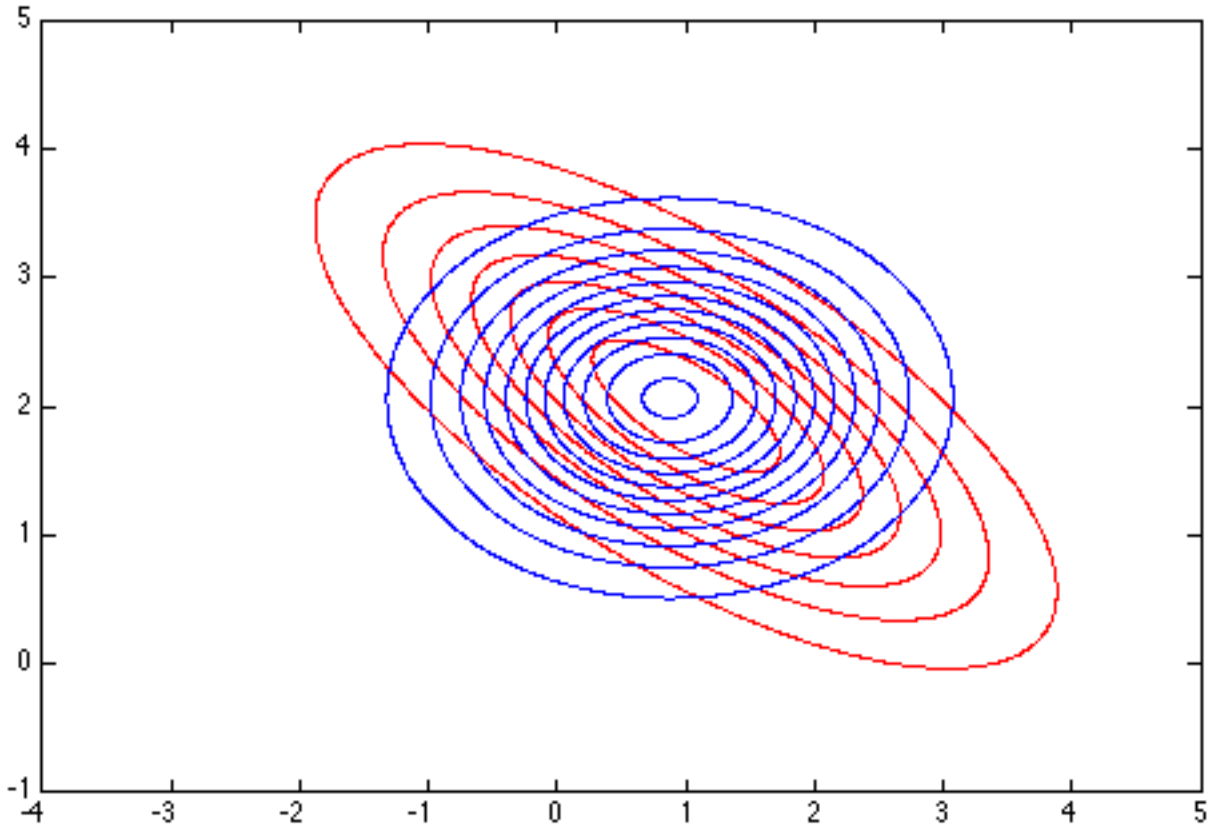
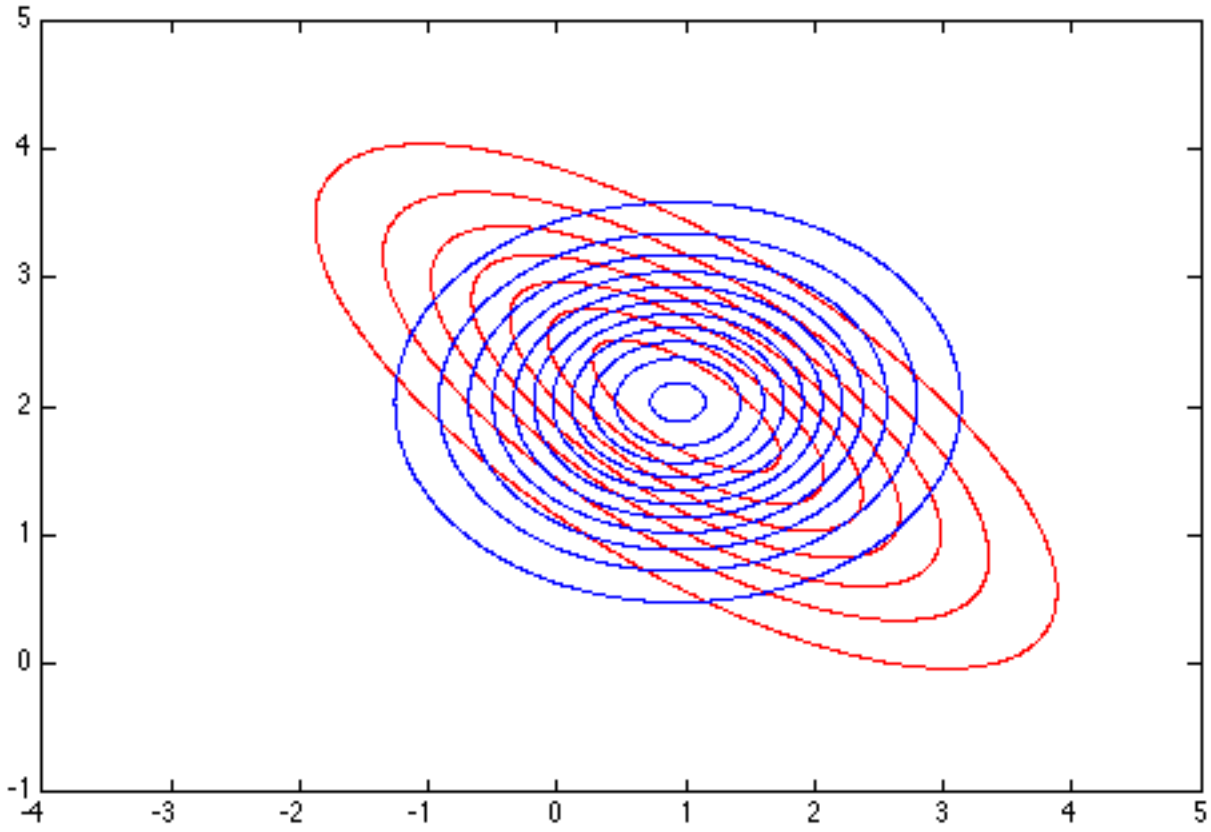
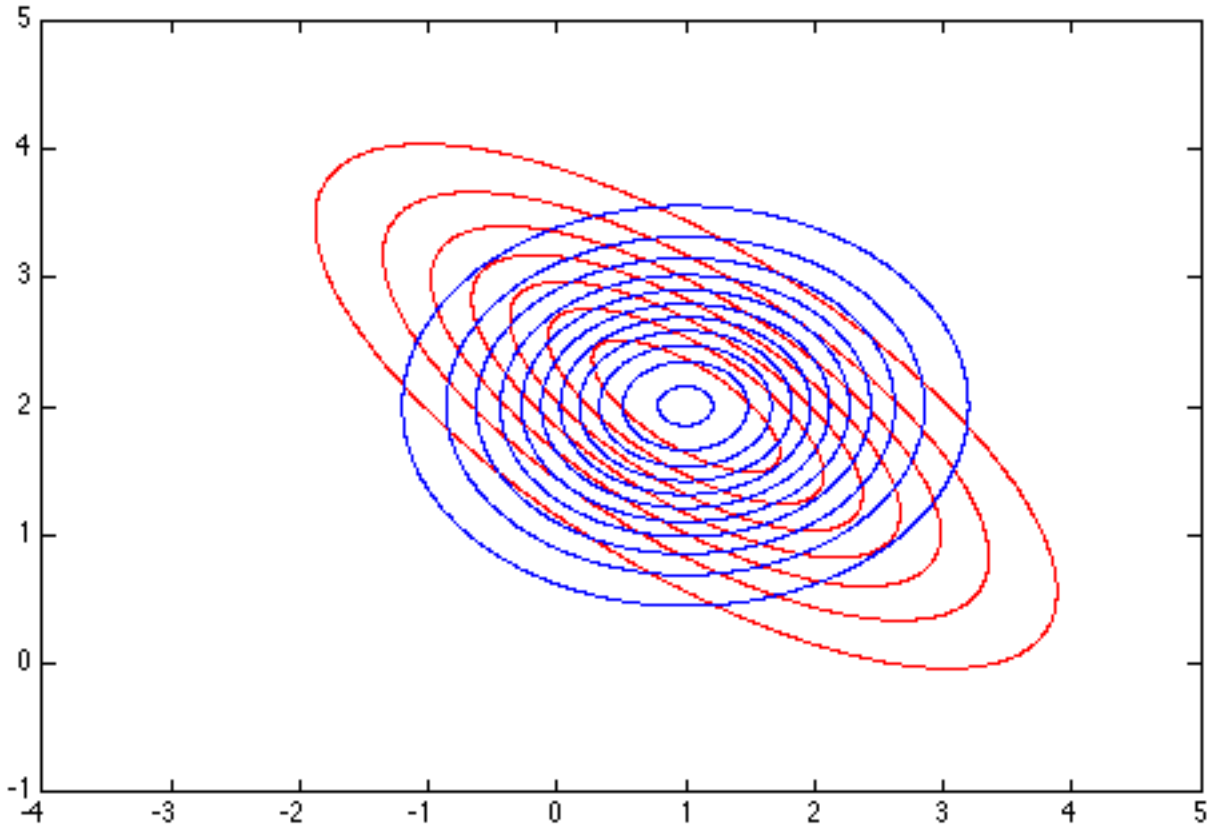


Variational Inference: Toy Example 1 – Approximating a Gaussian



Variational Inference: Toy Example 1 – Approximating a Gaussian



Variational Inference: Toy Example 1 – Approximating a Gaussian

Solving the linear system exactly, the joint variational distribution becomes:

$$q(\mathbf{z}) = q_1(z_1)q_2(z_2)$$

with parameters:

$$\boldsymbol{\mu}_q = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad \boldsymbol{\Lambda}_q = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

⇨ What happens if we minimize the reverse KL divergence *p* // *q* instead?

<u>Remember:</u>

$$p \| q \rightarrow \begin{cases} q_1^*(\mathbf{z}_j) = P(\mathbf{z}_1) \\ q_2^*(\mathbf{z}_j) = P(\mathbf{z}_2) \end{cases}$$

Variational Inference: Toy Example 1 – Approximating a Gaussian

Marginal of a multivariate Gaussian is easy:

$$q_1^*(z_1) = \mathcal{N}(z_1|m_1, \sigma_1^2) \qquad q_2^*(z_2) = \mathcal{N}(z_2|m_2, \sigma_2^2)$$

$$m_1 = \mu_1 \qquad m_2 = \mu_2$$

$$\sigma_1 = \Sigma_{11} \qquad \sigma_2 = \Sigma_{22}$$

where from before, we had:

$$\boldsymbol{\Sigma} = \begin{bmatrix} \Sigma_{11} & \Sigma_{21} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} = \boldsymbol{\Lambda}^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

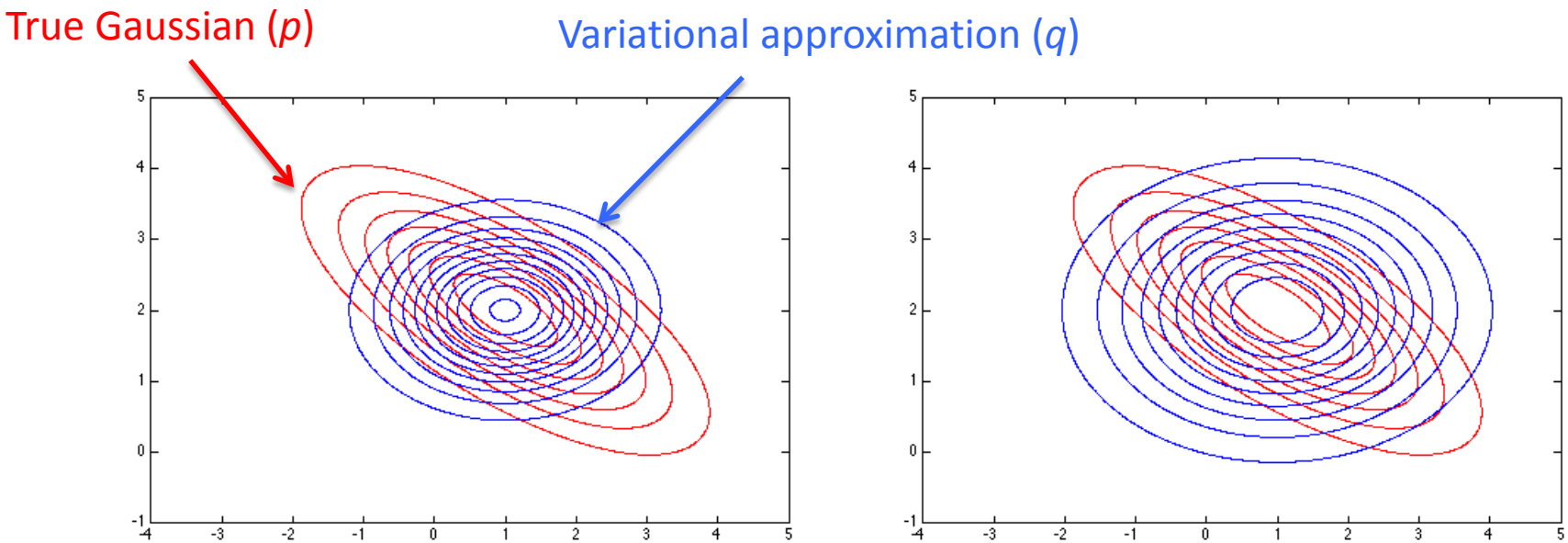
The joint variational distribution $q(z_1, z_2) = q_1(z_1)q_2(z_2)$ has parameters:

$$\boldsymbol{\mu} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad \boldsymbol{\Lambda}_q = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix}$$

Variational Inference: Toy Example 1 – Approximating a Gaussian

$$q||p \qquad\qquad\qquad\qquad p||q$$

$$\mathbf{\Lambda}_q = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \qquad\qquad \mathbf{\Lambda}_q = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix}$$

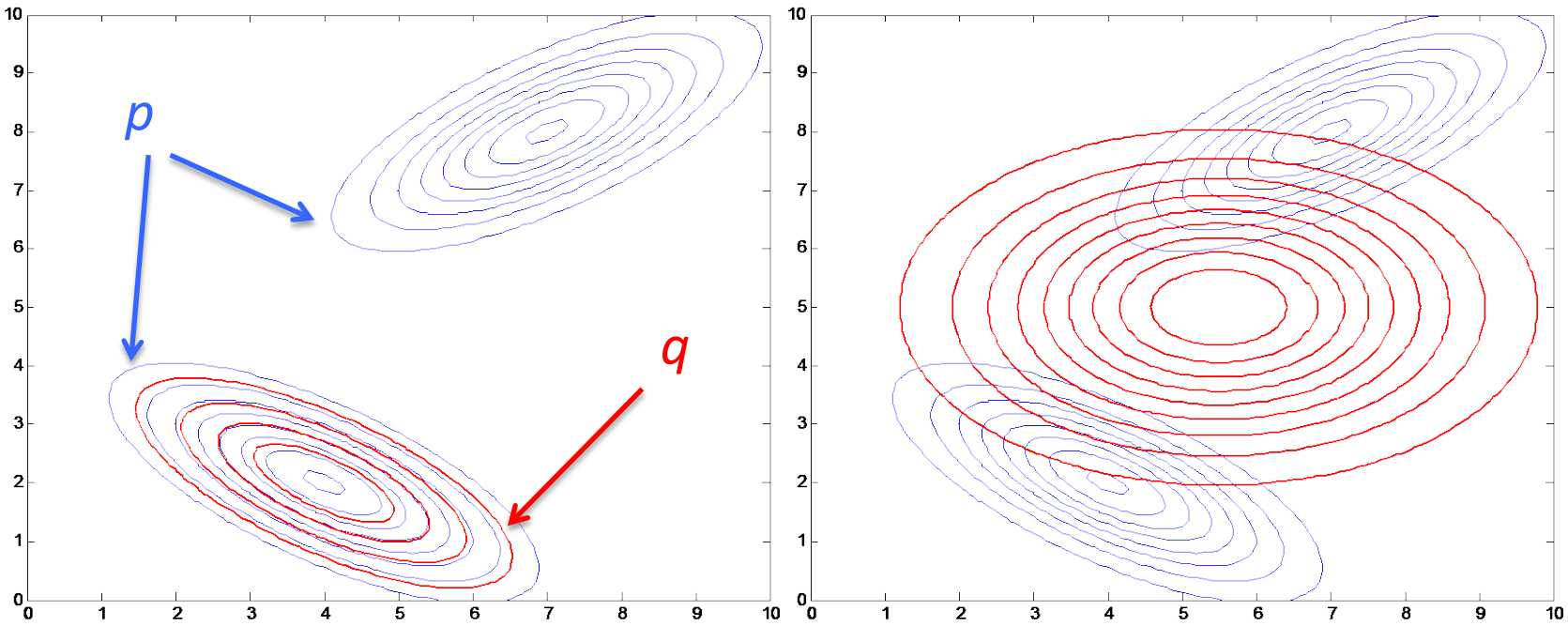


Univarite Gaussians Matlab code

Variational Inference: Toy Example 1 – Approximating a Gaussian

Univariate Variational approximation to a mixture of two Gaussians

$q||p$ $\qquad$ $p||q$

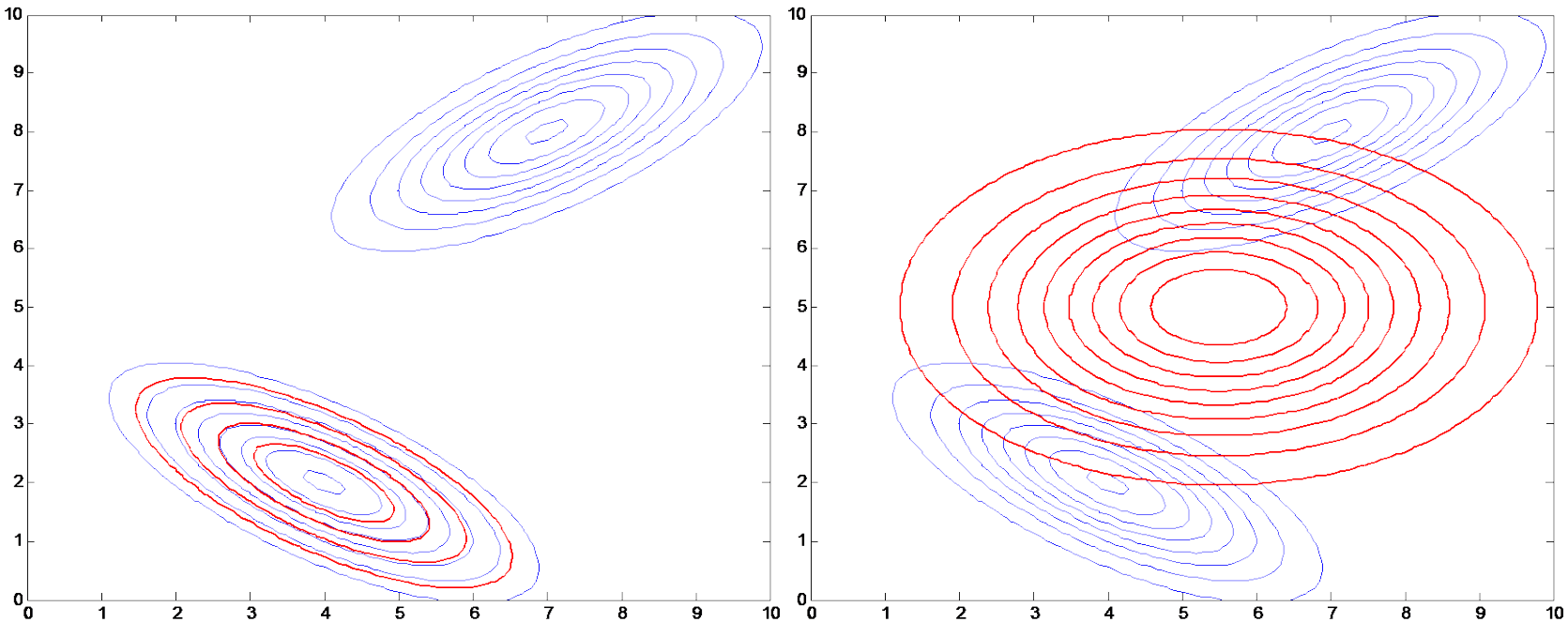


[Multimodal Gaussians Matlab code]

Variational Inference: Toy Example 1 – Approximating a Gaussian

Univariate Variational approximation to a mixture of two Gaussians

$$D_{\mathrm{KL}}(q\|p) = \int q(\mathbf{z}) \log\left(\frac{q(\mathbf{z})}{P(\mathbf{z})}\right) d\mathbf{z} \qquad D_{\mathrm{KL}}(p\|q) = \int P(\mathbf{z}) \log\left(\frac{P(\mathbf{z})}{q(\mathbf{z})}\right) d\mathbf{z}$$



Multimodal Gaussians Matlab code

Toy Example 2

Variational Inference: Toy Example 2 – Approximating a Posterior

Now we want to compute a posterior on the parameters, after seeing some data.

Our likelihood after *N* observations is given by:

$$P(\mathcal{D}|\mu,\tau) = \left(\frac{\tau}{2\pi}\right)^{N/2} \exp\left\{-\frac{\tau}{2}\sum_{n=1}^{N}(x_n-\mu)^2\right\}$$

Our prior on unknown mean and precision $p(\mu;\tau)$

$$P(\mu|\tau) = \mathcal{N}\left(\mu|\mu_0,(\lambda_0\tau)^{-1})\right)$$

$$P(\tau) = \mathrm{Gam}(\tau|\alpha_0,\beta_0)$$

We will assume a mean-field approximation for the posterior of the form:

$$q(\mu,\tau) = q_\mu(\mu)q_\tau(\tau)$$

Variational Inference: Toy Example 2 – Approximating a Posterior

As before, we minimize *q* // *p* KL to find optimal mean-field factors q_i :

$$\log q_\mu^*(\mu) = \mathbb{E}_\tau[\log P(\mathcal{D}, \mu, \tau)]$$
$$\log q_\tau^*(\tau) = \mathbb{E}_\mu[\log P(\mathcal{D}, \mu, \tau)]$$

Given our factorization of the joint distribution:

$$P(\mathcal{D}, \mu, \tau) = P(\mathcal{D}|\mu, \tau)P(\mu|\tau)P(\tau)$$

Our optimal mean-field factors q_i are:

$$\Longrightarrow \log q_\mu^*(\mu) = \mathbb{E}_\tau\left[\log P(\mathcal{D}|\mu, \tau) + \log P(\mu|\tau) + \log P(\tau)\right]$$
$$\log q_\tau^*(\tau) = \mathbb{E}_\mu\left[\log P(\mathcal{D}|\mu, \tau) + \log P(\mu|\tau) + \log P(\tau)\right]$$

Variational Inference: Toy Example 2 – Approximating a Posterior

Likelihood:

$$P(\mathcal{D}|\mu,\tau) = \left(\frac{\tau}{2\pi}\right)^{N/2} \exp\left\{-\frac{\tau}{2}\sum_{n=1}^{N}(x_n-\mu)^2\right\}$$

Prior

$$P(\mu|\tau) = \left(\frac{\lambda_0\tau}{2\pi}\right)^{1/2} \exp\left\{-\frac{\lambda_0\tau}{2}(\mu-\mu_0)^2\right\}$$

$$P(\tau) = \frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)}\tau^{\alpha_0-1}\exp\{-\beta_0\tau\}$$

$$\begin{aligned}
\log q_\mu^*(\mu) &= \mathbb{E}_\tau\left[\log P(\mathcal{D}|\mu,\tau) + \log P(\mu|\tau) + \log P(\tau)\right] \\
&= \mathbb{E}_\tau\left[\frac{N}{2}\log\frac{\tau}{2\pi} - \frac{\tau}{2}\sum^{N}(x_n-\mu)^2 + \frac{1}{2}\log\frac{\lambda_0\tau}{2\pi} - \frac{\lambda_0\tau}{2}(\mu-\mu_0)^2\right] \\
&= \mathbb{E}_\tau\left[-\frac{\tau}{2}\sum^{N}(x_n-\mu)^2 - \frac{\lambda_0\tau}{2}(\mu-\mu_0)^2\right] \\
&= -\frac{\mathbb{E}_\tau[\tau]}{2}\left(\sum^{N}(x_n-\mu)^2 + \lambda_0(\mu-\mu_0)^2\right)
\end{aligned}$$

Variational Inference: Toy Example 2 – Approximating a Posterior

$$\log q_\mu^*(\mu) = -\frac{\mathbb{E}_\tau[\tau]}{2}\left(\sum^N (x_n - \mu)^2 + \lambda_0(\mu - \mu_0)^2\right) + \text{const.}$$

The expression is quadratic in μ. Complete the square to get a Gaussian:

$$q_\mu^*(\mu) = \mathcal{N}(\mu|\tilde{\mu}, \tilde{\lambda}^{-1})$$

with parameters:

$$\tilde{\mu} = \frac{\lambda_0\mu_0 + N\bar{x}}{\lambda_0 + N}$$

$$\tilde{\lambda} = (\lambda_0 + N)\mathbb{E}[\tau]$$

Now, repeat the process with τ

Variational Inference: Toy Example 2 – Approximating a Posterior

Likelihood:

$$P(\mathcal{D}|\mu,\tau) = \left(\frac{\tau}{2\pi}\right)^{N/2} \exp\left\{-\frac{\tau}{2}\sum_{n=1}^{N}(x_n-\mu)^2\right\}$$

Prior

$$\begin{cases} P(\mu|\tau) = \left(\dfrac{\lambda_0\tau}{2\pi}\right)^{1/2} \exp\left\{-\dfrac{\lambda_0\tau}{2}(\mu-\mu_0)^2\right\} \\ P(\tau) = \dfrac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)}\tau^{\alpha_0-1}\exp\{-\beta_0\tau\} \end{cases}$$

$$\log q_\tau^*(\tau) = \mathbb{E}_\mu\left[\log P(\mathcal{D}|\mu,\tau) + \log P(\mu|\tau) + \log P(\tau)\right]$$

$$\mathbb{E}_\mu[\log P(\mathcal{D}|\mu,\tau)] = \mathbb{E}_\mu\left[\frac{N}{2}\log\frac{\tau}{2\pi} - \frac{\tau}{2}\sum^{N}(x_n-\mu)^2\right] = \frac{N}{2}\log\tau - \frac{\tau}{2}\mathbb{E}_\mu\left[\sum^{N}(x_n-\mu)^2\right] + \mathrm{C}$$

$$\mathbb{E}_\mu[\log P(\mu|\tau)] = \mathbb{E}_\mu\left[\frac{1}{2}\log\frac{\tau\lambda_0}{2\pi} - \frac{\lambda_0\tau}{2}(\mu-\mu_0)^2\right] = \frac{1}{2}\log\tau - \frac{\lambda_0\tau}{2}\mathbb{E}_\mu(\mu-\mu_0)^2 + C$$

$$\mathbb{E}_\mu[\log P(\tau)] = \mathbb{E}_\mu\left[(\alpha_0-1)\tau - \beta_0\tau\right] = (\alpha_0-1)\tau - \beta_0\tau + C$$

$$\boxed{\log q_\tau^*(\tau) = -\frac{1}{2}\mathbb{E}_\mu\left[\sum^{N}(x_n-\mu)^2 + \lambda_0(\mu-\mu_0)^2\right] + \frac{1}{2}\log\tau(N+1) + (\alpha_0-\beta_0-1)\tau + C}$$

Variational Inference: Toy Example 2 – Approximating a Posterior

$$\log q_\tau^*(\tau) = -\frac{1}{2}\mathbb{E}_\mu\left[\sum^N (x_n-\mu)^2 + \lambda_0(\mu-\mu_0)^2\right] + \frac{1}{2}\log\tau(N+1) + (\alpha_0-\beta_0-1)\tau + C$$

We recognize that as the gamma distribution with parameters:

$$\tilde{\alpha} = \alpha_0 + \frac{N+1}{2}$$

$$\tilde{\beta} = \beta_0 + \frac{1}{2}\mathbb{E}_\mu\underbrace{\left[\sum^N (x_n-\mu)^2 + \lambda_0(\mu-\mu_0)^2\right]}$$

Finally, expanding the last expectation:

$$\frac{1}{2}\mathbb{E}_\mu\left[\sum^N (x_n-\mu)^2 + \lambda_0(\mu-\mu_0)^2\right] = \frac{1}{2}\left[\left(\sum^N x_n^2 - 2\mathbb{E}[\mu]\sum^N x_n + N\mathbb{E}[\mu^2]\right) + \lambda_0\mathbb{E}[\mu^2] - 2\lambda_0\mu_0\mathbb{E}[\mu]\right]$$

$$= \underbrace{\frac{1}{2}\left[\mathbb{E}[\mu^2](N+\lambda_0) - 2\mathbb{E}[\mu](\sum^N x_n + \lambda_0\mu_0) + \sum^N x_n^2\right]}_{\text{Coupled Expectations as before}}$$

Variational Inference: Toy Example 2 – Approximating a Posterior

Final parameter update equations:

$$\tilde{\alpha} = \alpha_0 + \frac{N+1}{2}$$

$$\tilde{\beta} = \beta_0 + \frac{1}{2}\left[\underline{\mathbb{E}[\mu^2]}(N+\lambda_0) - 2\underline{\mathbb{E}[\mu]}\left(\sum^N x_n + \lambda_0\mu_0\right) + \sum^N x_n^2\right]$$

$$\tilde{\mu} = \frac{\lambda_0\mu_0 + N\bar{x}}{\lambda_0 + N}$$

$$\tilde{\lambda} = (\lambda_0 + N)\underline{\mathbb{E}[\tau]}$$

Useful facts:

$$\mathbb{E}[\mu] = \tilde{\mu}$$

$$\mathbb{E}[\mu^2] = \tilde{\mu}^2 + \tilde{\lambda}^{-1}$$

$$\mathbb{E}[\tau] = \frac{\tilde{\alpha}}{\tilde{\beta}}$$

Variational Inference: Toy Example 2 – Approximating a Posterior

As before, we can write down an iterative algorithm:

Initialize:

$$\tilde{\alpha} \leftarrow rand \quad \tilde{\beta} \leftarrow rand \quad \tilde{\lambda} \leftarrow rand \quad \tilde{\mu} \leftarrow rand$$

While not converged:

$$\mathbb{E}[\tau] \leftarrow \frac{\tilde{\alpha}}{\tilde{\beta}}$$

$$\tilde{\mu} \leftarrow \frac{\lambda_0 \mu_0 + N\bar{x}}{\lambda_0 + N}$$

$$\tilde{\lambda} \leftarrow (\lambda_0 + N)\mathbb{E}[\tau]$$

$$\mathbb{E}[\mu] \leftarrow \tilde{\mu}$$

$$\mathbb{E}[\mu^2] \leftarrow \tilde{\mu}^2 + \tilde{\lambda}^{-1}$$

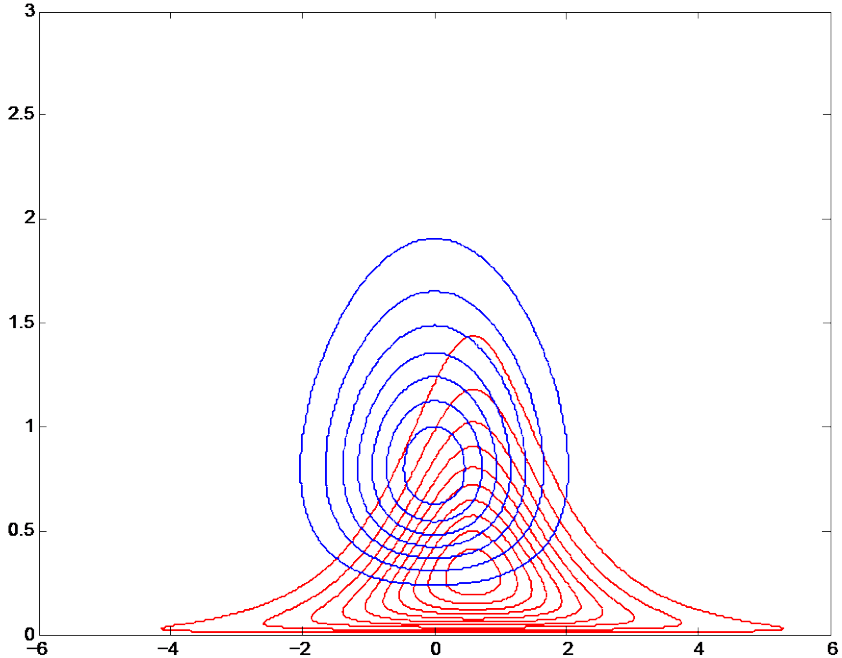
$$\tilde{\alpha} \leftarrow \alpha_0 + \frac{N+1}{2}$$

$$\tilde{\beta} \leftarrow \beta_0 + \frac{1}{2}\left[\mathbb{E}[\mu^2](N + \lambda_0) - 2\mathbb{E}[\mu](\sum^N x_n + \lambda_0\mu_0) + \sum^N x_n^2\right]$$

Can also be solved without iteration

Variational Inference: Toy Example 2 – Approximating a Posterior

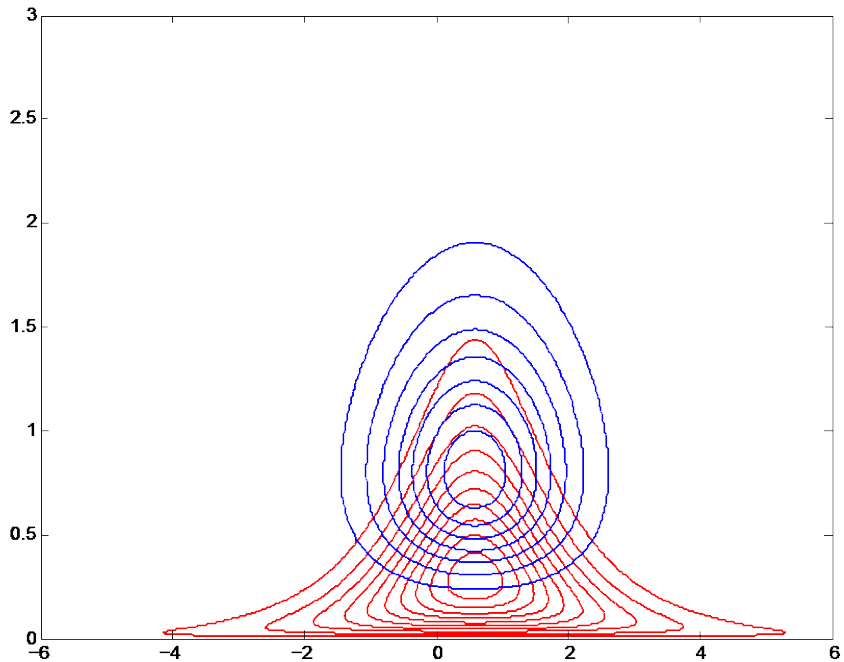
Initialization



Matlab code

Variational Inference: Toy Example 2 – Approximating a Posterior

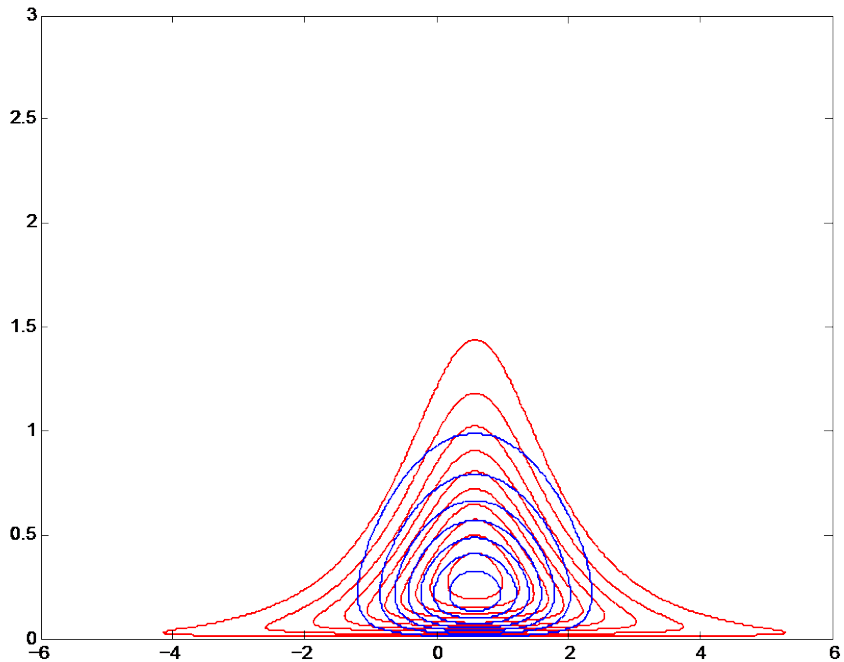
μ converged



Matlab code

Variational Inference: Toy Example 2 – Approximating a Posterior

μ, λ converged

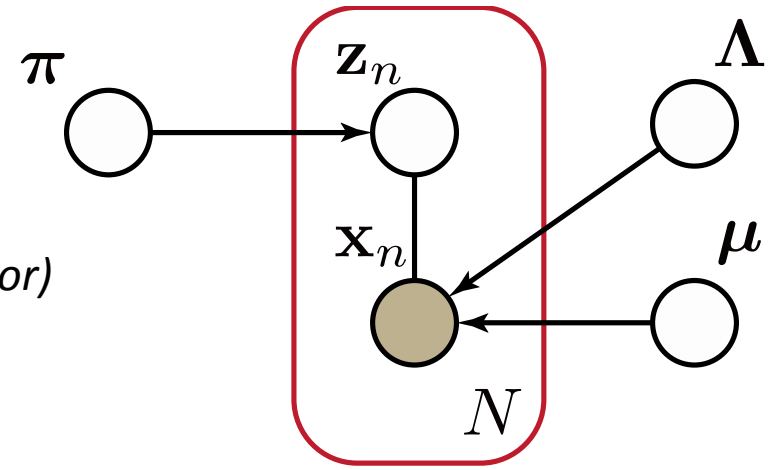


Matlab code

Gaussian Mixture

Variational Inference: Gaussian Mixture Model

<u>A classical mixture model</u>:



Observations: $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$

Latent variables: $\mathbf{Z} = \{\mathbf{z}_1, \ldots, \mathbf{z}_N\}$ *(binary vector)*

Conditional distributions:

$$P(\mathbf{X}|\mathbf{Z}, \boldsymbol{\mu}, \boldsymbol{\Lambda}) = \prod_{n=1}^{N} \prod_{k=1}^{K} \mathcal{N}\left(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k^{-1}\right)^{z_{nk}}$$

$$P(\mathbf{Z}|\boldsymbol{\pi}) = \prod_{n=1}^{N} \prod_{k=1}^{K} \pi_k^{z_{nk}}$$

$$P(\boldsymbol{\pi}) = \mathrm{Dir}(\boldsymbol{\pi}|\boldsymbol{\alpha}_0) = C(\boldsymbol{\alpha}_0) \prod_{k=1}^{K} \pi_k^{\alpha_0 - 1}$$

$$P(\boldsymbol{\mu}, \boldsymbol{\Lambda}) = P(\boldsymbol{\mu}, \boldsymbol{\Lambda}) P(\boldsymbol{\Lambda})$$

$$= \prod_{k=1}^{K} \mathcal{N}\left(\boldsymbol{\mu}_k | \mathbf{m}_0, (\beta_0 \boldsymbol{\Lambda}_k)^{-1}\right) \mathcal{W}(\boldsymbol{\Lambda}_k | \boldsymbol{W}_0, \nu_0)$$

$$P(\mathbf{X}, \mathbf{Z}, \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda}) = p(\mathbf{X}|\mathbf{Z}, \boldsymbol{\mu}, \boldsymbol{\Lambda}) P(\mathbf{Z}|\boldsymbol{\pi}) P(\boldsymbol{\pi}) P(\boldsymbol{\mu}|\boldsymbol{\Lambda}) P(\boldsymbol{\Lambda})$$

Variational Inference: Gaussian Mixture Model

$$P(\mathbf{X}, \mathbf{Z}, \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda}) = p(\mathbf{X}|\mathbf{Z}, \boldsymbol{\mu}, \boldsymbol{\Lambda})P(\mathbf{Z}|\boldsymbol{\pi})P(\boldsymbol{\pi})P(\boldsymbol{\mu}|\boldsymbol{\Lambda})P(\boldsymbol{\Lambda})$$

We want to compute the posterior $P(\mathbf{Z}, \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda}|\mathbf{X})$

Difficult to compute exactly. Propose an approximate distribution of the form:

$$q(\mathbf{Z}, \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda}) = q(\mathbf{Z})q(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda})$$

With no further assumptions, this is sufficient to obtain a tractable posterior

Exactly as before, find the optimal q_i by minimizing q // p KL:

$$\Longrightarrow \quad \log q^*(\mathbf{Z}) = \mathbb{E}_{\boldsymbol{\pi},\boldsymbol{\mu},\boldsymbol{\Lambda}}[\log P(\mathbf{X}, \mathbf{Z}, \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda}] + C$$

$$\log q^*(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda}) = \mathbb{E}_{\mathbf{Z}}[\log P(\mathbf{X}, \mathbf{Z}, \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda})] + C$$

Variational Inference: Gaussian Mixture Model

$$\log q^*(\mathbf{Z}) = \mathbb{E}_{\boldsymbol{\pi},\boldsymbol{\mu},\boldsymbol{\Lambda}}[\log P(\mathbf{X}, \mathbf{Z}, \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda}] + C$$

Plug in the factorized form of the joint distribution from before:

$$\log q^*(\mathbf{Z}) = \mathbb{E}_{\boldsymbol{\pi}}[\log P(\mathbf{Z}|\boldsymbol{\pi})] + \mathbb{E}_{\boldsymbol{\mu},\boldsymbol{\Lambda}}[\log P(\mathbf{X}|\mathbf{Z}, \boldsymbol{\mu}, \boldsymbol{\Lambda})]$$

$$\log q^*(\mathbf{Z}) = \sum^N \sum^K z_{nk} \underbrace{\left(\mathbb{E}[\log \pi_k] + \frac{1}{2}\mathbb{E}[\log |\boldsymbol{\Lambda_k}|] - \frac{D}{2}\log(2\pi) - \frac{1}{2}\mathbb{E}_{\boldsymbol{\mu}_k,\boldsymbol{\Lambda}_k}[(\mathbf{x}_n - \boldsymbol{\mu}_k)^T \boldsymbol{\Lambda_k}(\mathbf{x}_n - \boldsymbol{\mu}_k)]\right)}_{\rho_{nk}}$$

Taking the exponent of the stuff inside the parenthesis returns back the original distribution with updated parameters:

$$q^*(\mathbf{Z}) = \prod^N \prod^K r_{nk}^{z_{nk}} \quad \text{where} \quad r_{nk} = \frac{\rho_{nk}}{\sum^K \rho_{nk}}$$

Variational Inference: Gaussian Mixture Model

$$\log q^*(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda}) = \mathbb{E}_{\mathbf{Z}}[\log P(\mathbf{X}, \mathbf{Z}, \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda})] + C$$

As before, plug in the factorization for the joint distribution:

$$\log q^*(\boldsymbol{\pi}, \boldsymbol{\lambda}, \boldsymbol{\Lambda}) = \log P(\boldsymbol{\pi}) + \sum_{k=1}^{K} \log P(\boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k) + \mathbb{E}_{\mathbf{Z}}[\log P(\mathbf{Z}|\boldsymbol{\pi})]$$

$$\ldots \quad + \sum_{n=1}^{N} \sum_{k=1}^{K} \mathbb{E}[z_{nk}] \log \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k^{-1}) + C$$

Observe that we can partition the variables above to factorize further:

$$q(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Lambda}) = q(\boldsymbol{\pi}) \prod_{k=1}^{K} q(\boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k)$$

Variational Inference: Gaussian Mixture Model

$$\log q^*(\boldsymbol{\pi}, \boldsymbol{\lambda}, \boldsymbol{\Lambda}) = \underline{\log P(\boldsymbol{\pi})} + \sum_{k=1}^{K} \log P(\boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k) + \underline{\mathbb{E}_{\mathbf{Z}}[\log P(\mathbf{Z}|\boldsymbol{\pi})]}$$

$$\ldots \quad + \sum_{n=1}^{N} \sum_{k=1}^{K} \mathbb{E}[z_{nk}] \log \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k^{-1}) + C$$

Considering only the terms that depend on : $\boldsymbol{\pi}$

$$q^*(\boldsymbol{\pi}) = (\alpha_0 - 1) \sum_{k=1}^{K} \log \pi_k + \sum_{k=1}^{K} \sum_{n=1}^{N} r_{nk} \log \pi_k + C$$

This is an Dirichlet distribution:

$$q^*(\boldsymbol{\pi}) = \mathrm{Dir}(\boldsymbol{\pi} | \boldsymbol{\alpha})$$

with updated parameters:

$$\alpha_k = \alpha_0 + \sum_{n=1}^{N} r_{nk}$$

Variational Inference: Gaussian Mixture Model

$$\log q^*(\boldsymbol{\pi}, \boldsymbol{\lambda}, \boldsymbol{\Lambda}) = \log P(\boldsymbol{\pi}) + \underline{\sum_{k=1}^{K} \log P(\boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k)} + \mathbb{E}_{\mathbf{Z}}[\log P(\mathbf{Z}|\boldsymbol{\pi})]$$

$$\ldots \quad + \underline{\sum_{n=1}^{N} \sum_{k=1}^{K} \mathbb{E}[z_{nk}] \log \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k^{-1})} + C$$

Considering only the terms that depend on $\boldsymbol{\mu}, \boldsymbol{\Lambda}$, we recognize:

$$q^*(\boldsymbol{\mu}_k, \boldsymbol{\Lambda}_k) = \mathcal{N}\left(\boldsymbol{\mu}_k | \mathbf{m}_k, (\beta_k \boldsymbol{\Lambda}_k)^{-1}\right) \mathcal{W}(\boldsymbol{\Lambda}_k, \nu_k)$$

This is an updated Normal-Wishart distribution with new parameters:

$$\beta_k = \beta_0 + N_k$$

$$\mathbf{m}_k = \frac{1}{\beta_k}(\beta_0 \mathbf{m}_0 + N_k \bar{\mathbf{x}}_k)$$

$$\mathbf{W}_k^{-1} = \mathbf{W}_0^{-1} + \mathbf{S}_k + \frac{\beta_0 N_k}{\beta_0 + N_k}(\bar{\mathbf{x}}_k - \mathbf{m}_0)(\bar{\mathbf{x}}_k - \mathbf{m}_0)^T$$

$$\nu_k = \nu_0 + N_k + 1$$

Variational Inference: Gaussian Mixture Model

Updated Normal-Wishart parameters:

$$\beta_k = \beta_0 + N_k$$

$$\mathbf{m}_k = \frac{1}{\beta_k}(\beta_0 \mathbf{m}_0 + N_k \bar{\mathbf{x}}_k)$$

$$\mathbf{W}_k^{-1} = \mathbf{W}_0^{-1} + \mathbf{S}_k + \frac{\beta_0 N_k}{\beta_0 + N_k}(\bar{\mathbf{x}}_k - \mathbf{m}_0)(\bar{\mathbf{x}}_k - \mathbf{m}_0)^T$$

$$\nu_k = \nu_0 + N_k + 1$$

where we defined:

$$N_k = \sum_{n=1}^{N} r_{nk}$$

$$\bar{\mathbf{x}}_k = \frac{1}{N_k} \sum_{n=1}^{N} r_{nk} \mathbf{x}_n$$

$$\mathbf{S}_k = \frac{1}{N_k} \sum_{n=1}^{N} r_{nk} (\mathbf{x}_n - \bar{\mathbf{x}}_k)(\mathbf{x}_n - \bar{\mathbf{x}}_k)^T$$

Variational Inference: Gaussian Mixture Model

Dirichlet and Normal-Wishart updates, both require r_{nk}

To compute r_{nk} we need to compute a number of expectations as before.

Recall:

$$\underbrace{\left(\mathbb{E}[\log \pi_k] + \frac{1}{2}\mathbb{E}[\log |\boldsymbol{\Lambda_k}|] - \frac{D}{2}\log(2\pi) - \frac{1}{2}\mathbb{E}_{\boldsymbol{\mu}_k,\boldsymbol{\Lambda}_k}[(\mathbf{x}_n - \boldsymbol{\mu}_k)^T \boldsymbol{\Lambda_k}(\mathbf{x}_n - \boldsymbol{\mu}_k)]\right)}_{\rho_{nk}}$$

where $r_{nk} = \dfrac{\rho_{nk}}{\sum^K \rho_{nk}}$

Thus we need to compute based on the current distributions:

$$\mathbb{E}_{\boldsymbol{\mu}_k,\boldsymbol{\Lambda}_k}[(\mathbf{x}_n - \boldsymbol{\mu}_k)^T \boldsymbol{\Lambda}_k(\mathbf{x}_n - \boldsymbol{\mu}_k)]$$

$$\mathbb{E}[\log |\boldsymbol{\Lambda}_k|]$$

$$\mathbb{E}[\log \pi_k]$$

Variational Inference: Gaussian Mixture Model

The expectations are computed using properties of Normal and Wishart pdfs:

$$\mathbb{E}_{\boldsymbol{\mu}_k,\boldsymbol{\Lambda}_k}[(\mathbf{x}_n - \boldsymbol{\mu}_k)^T \boldsymbol{\Lambda}_k (\mathbf{x}_n - \boldsymbol{\mu}_k)] = D\beta_k^{-1} + \nu_k (\mathbf{x}_n - \mathbf{m}_n)^T \mathbf{W}_k (\mathbf{x}_n - \mathbf{m}_k)$$

$$\mathbb{E}[\log|\boldsymbol{\Lambda}_k|] = \sum_{i=1}^{D} \psi\left(\frac{\nu_k + 1 - i}{2}\right) + D\log 2 + \log|\mathbf{W}_k|$$

$$\mathbb{E}[\log|\boldsymbol{\pi}_k|] = \psi(\alpha_k) - \psi\left(\sum^{K} \alpha_k\right)$$

Substituting these results back into the ρ_{nk} (un-normalized responsibilities):

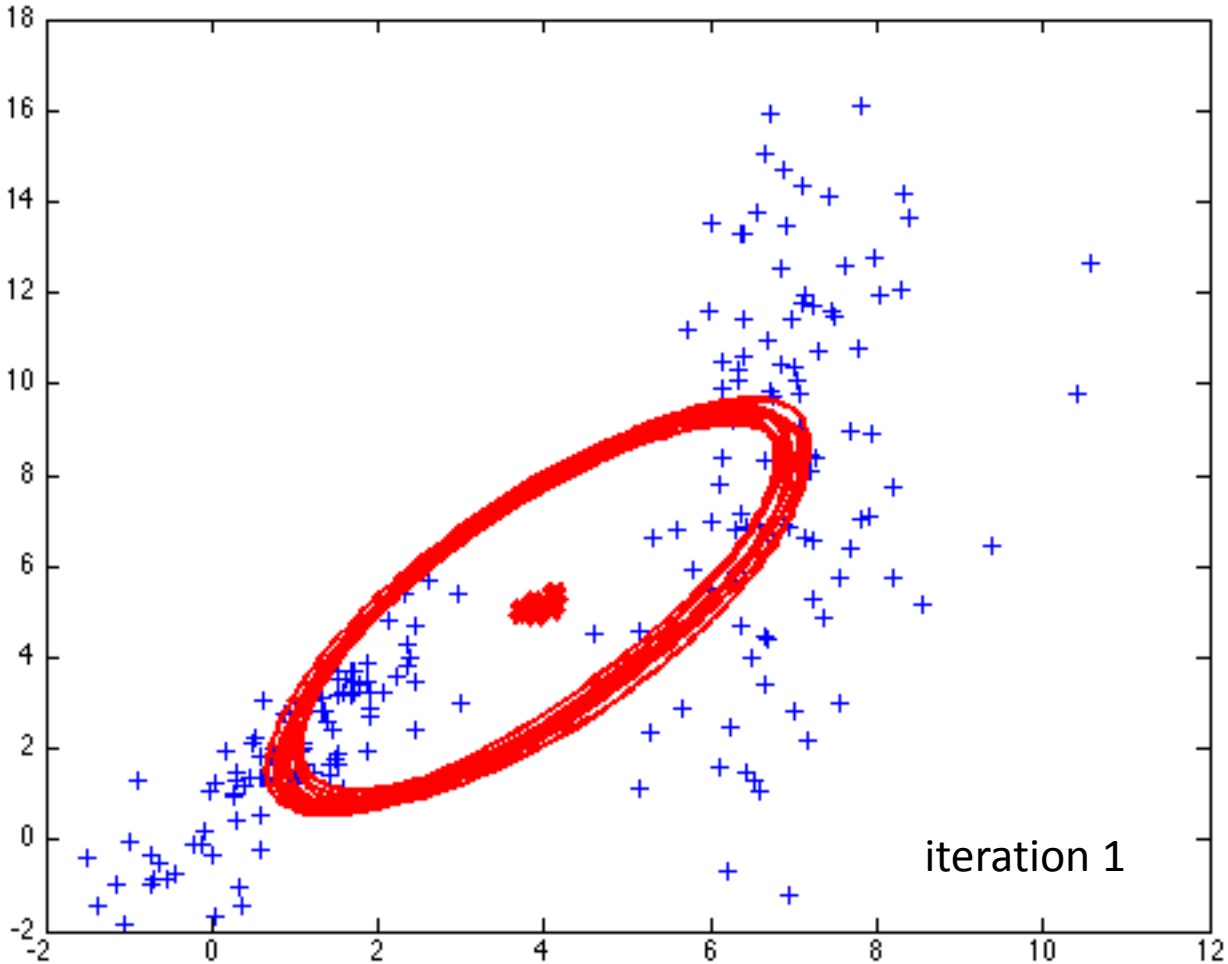
$$\rho_{nk} \propto \exp\left\{-\frac{D}{2\beta_k} - \frac{\nu_k}{2}(\mathbf{x}_n - \mathbf{m}_k)^T \mathbf{W}_k (\mathbf{x}_n - \mathbf{m}_k)\right\}$$

Expectation Maximization yields a similar result for responsibilities:

$$\rho_{nk} \propto \exp\left\{-\frac{1}{2}(\mathbf{x}_n - \boldsymbol{\mu}_k)^T \boldsymbol{\Lambda}_k (\mathbf{x}_n - \boldsymbol{\mu}_k)\right\}$$

Variational Inference: Gaussian Mixture Model

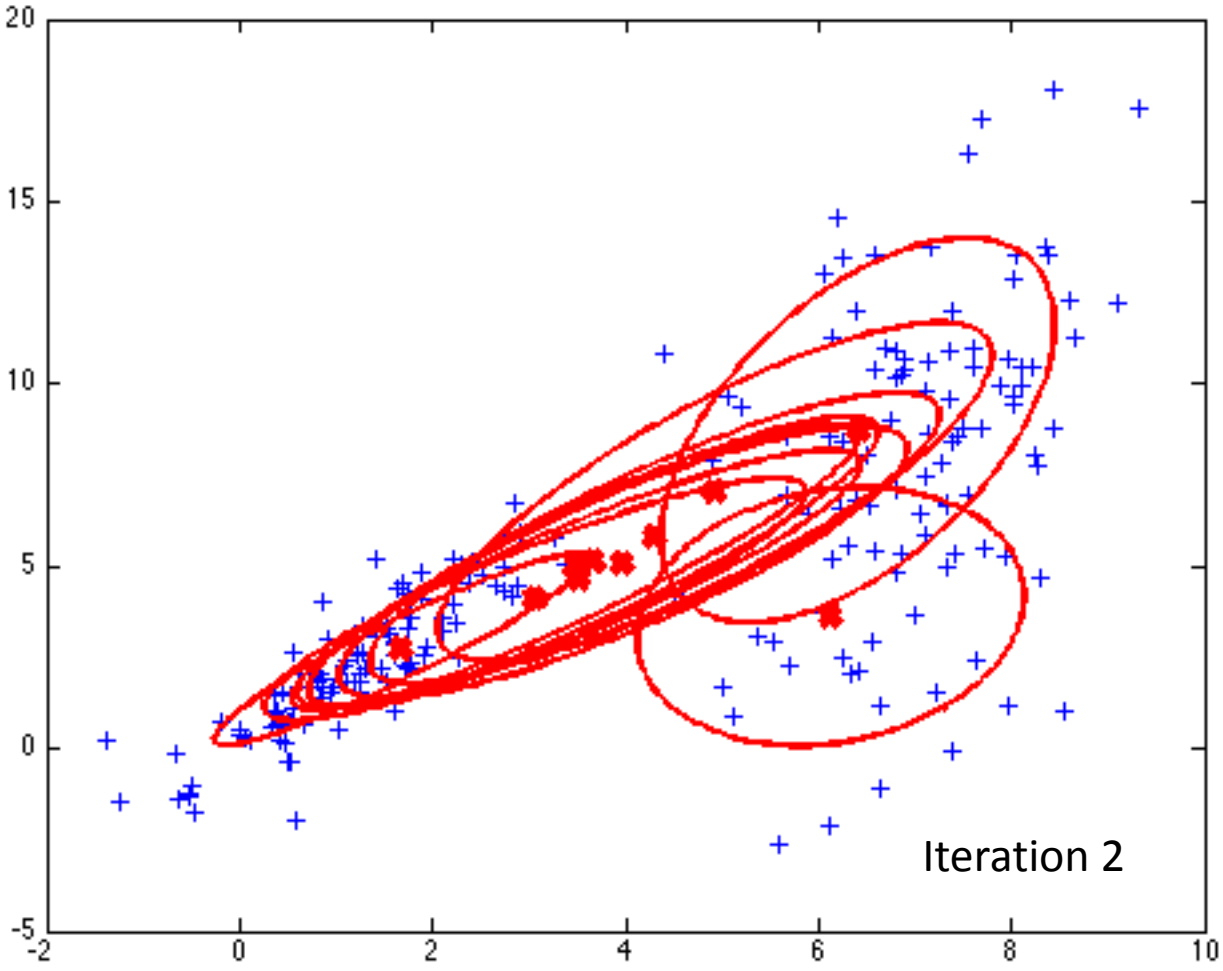
Variational Bayes Gaussian Mixture. (K=10, 2 true Gaussians)



Matlab code

Variational Inference: Gaussian Mixture Model

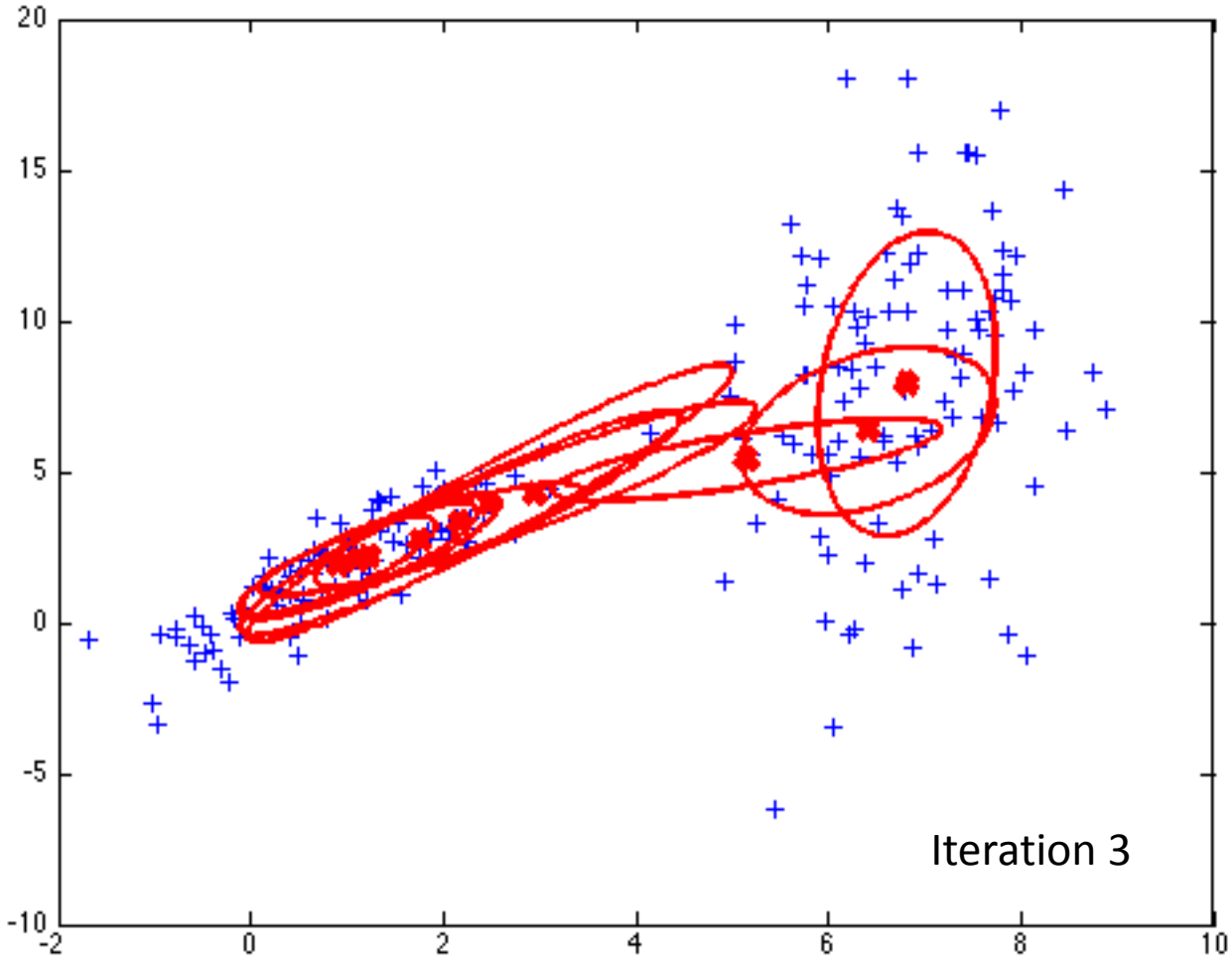
Variational Bayes Gaussian Mixture. (K=10, 2 true Gaussians)



Matlab code

Variational Inference: Gaussian Mixture Model

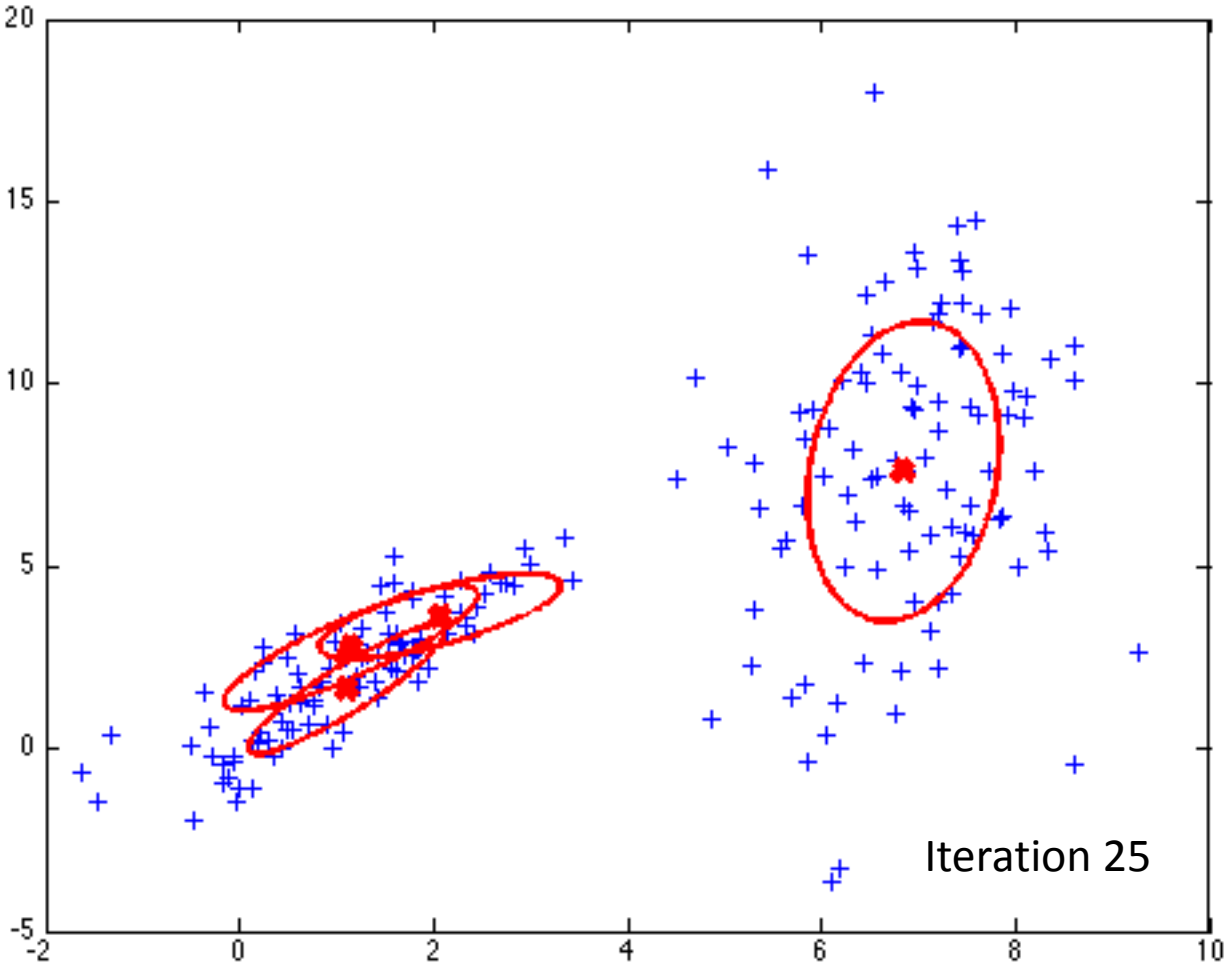
Variational Bayes Gaussian Mixture. (K=10, 2 true Gaussians)



Matlab code

Variational Inference: Gaussian Mixture Model

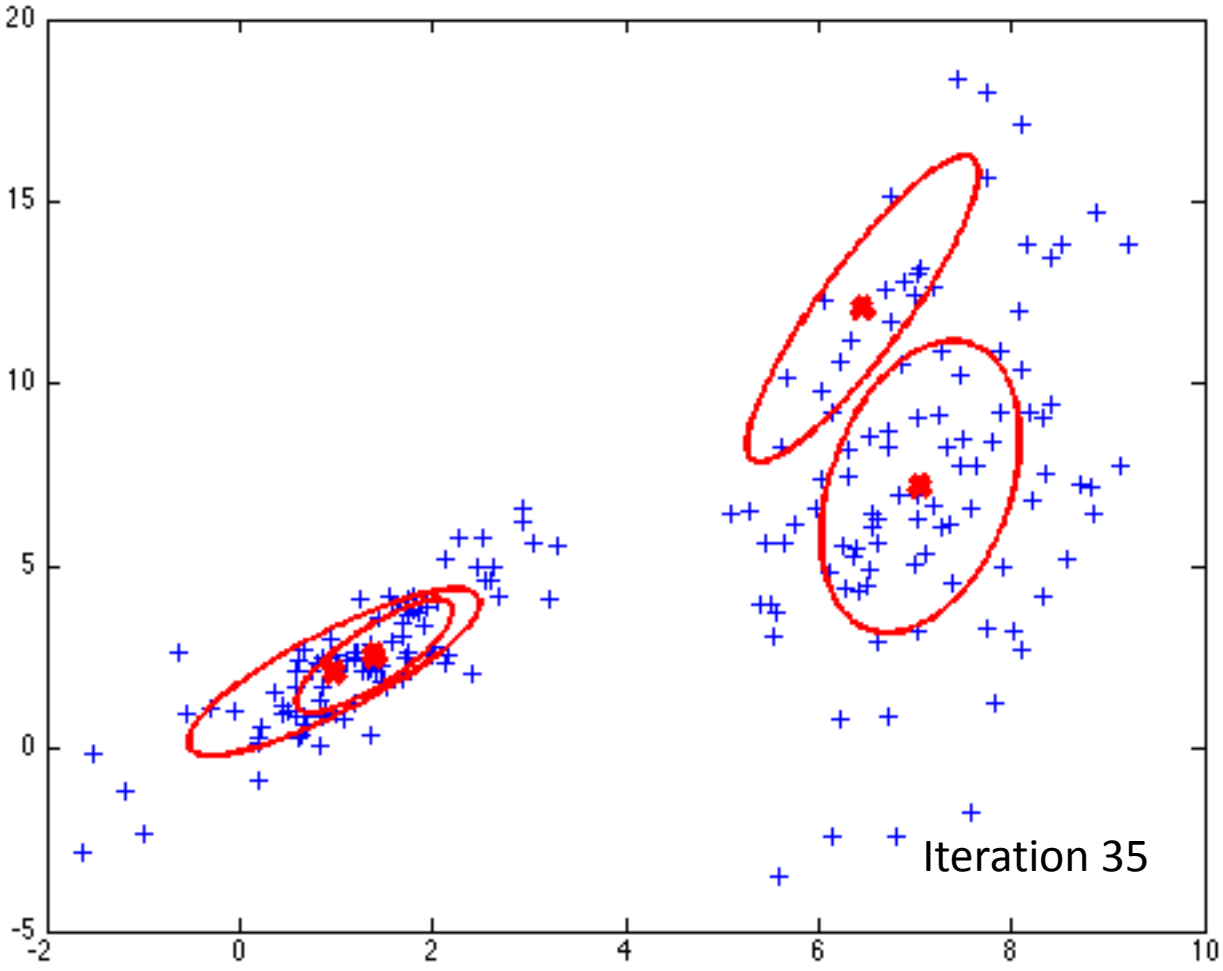
Variational Bayes Gaussian Mixture. (K=10, 2 true Gaussians)



Matlab code

Variational Inference: Gaussian Mixture Model

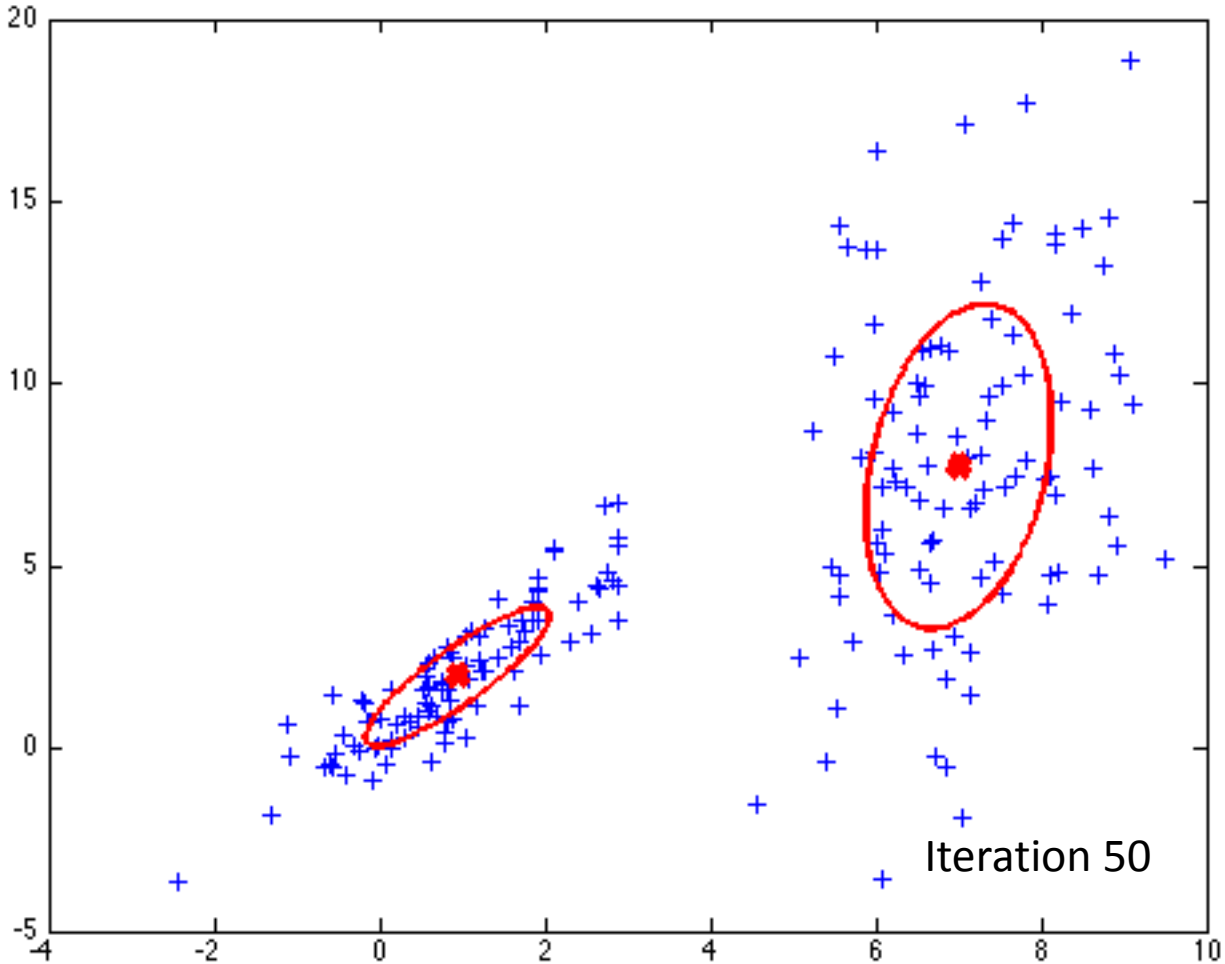
Variational Bayes Gaussian Mixture. (K=10, 2 true Gaussians)



Matlab code

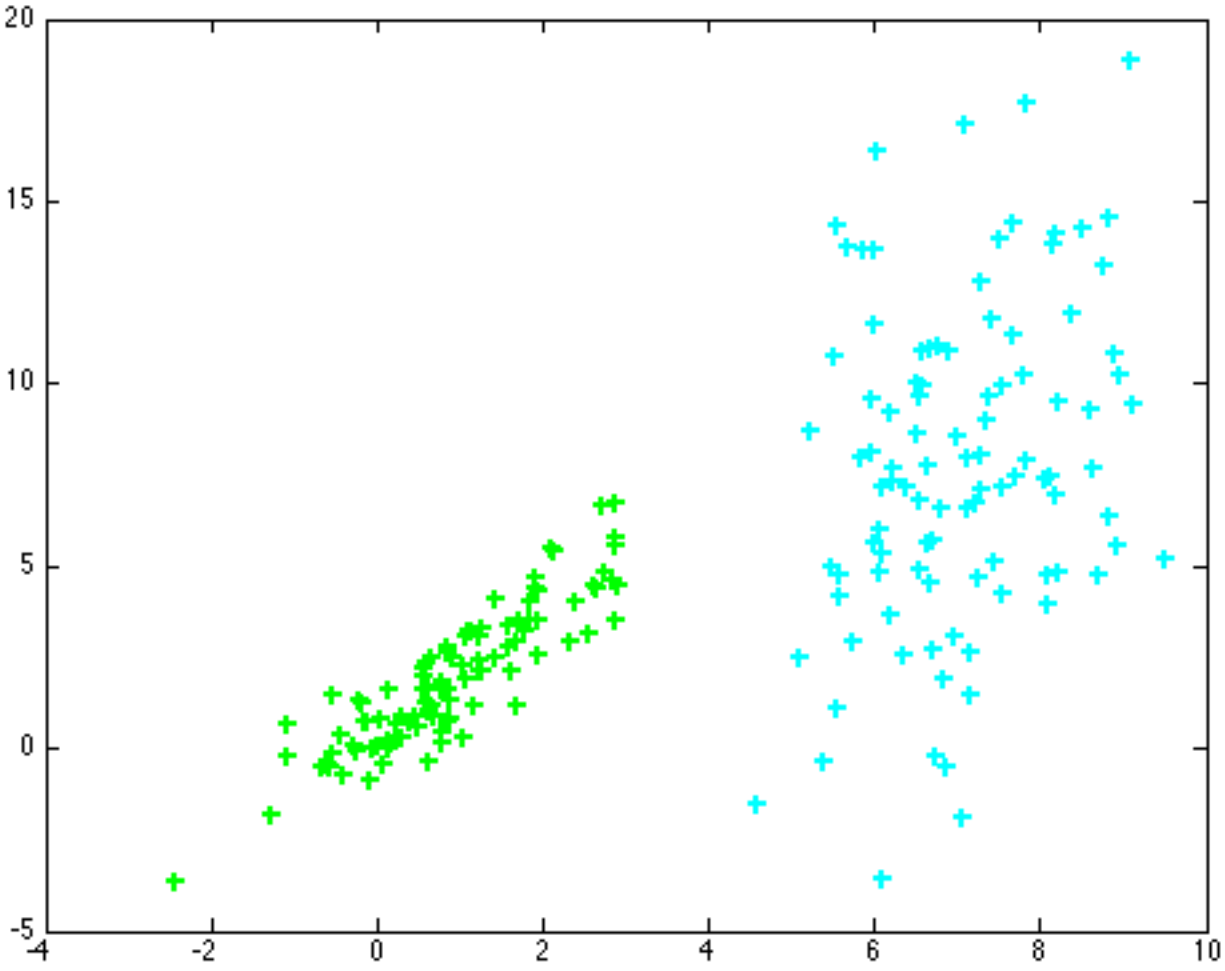
Variational Inference: Gaussian Mixture Model

Variational Bayes Gaussian Mixture. (K=10, 2 true Gaussians)



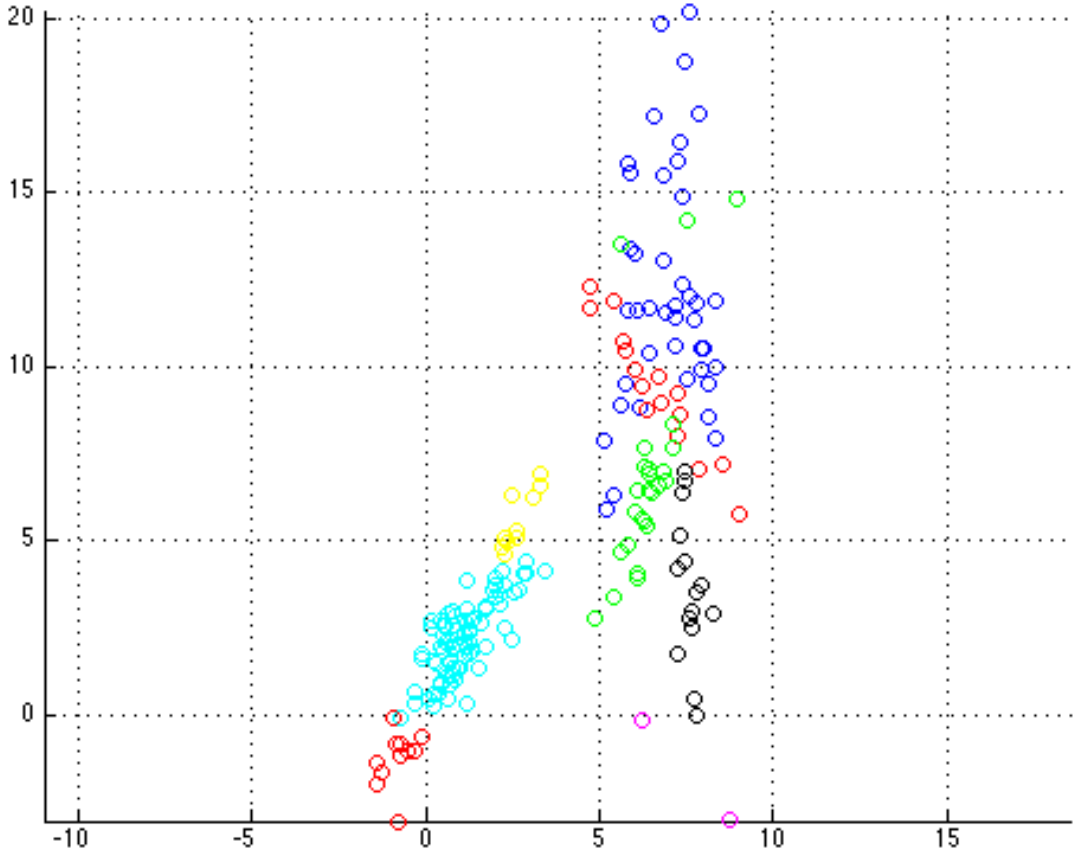
Matlab code

Variational Inference: Gaussian Mixture Model



Variational Inference: Gaussian Mixture Model

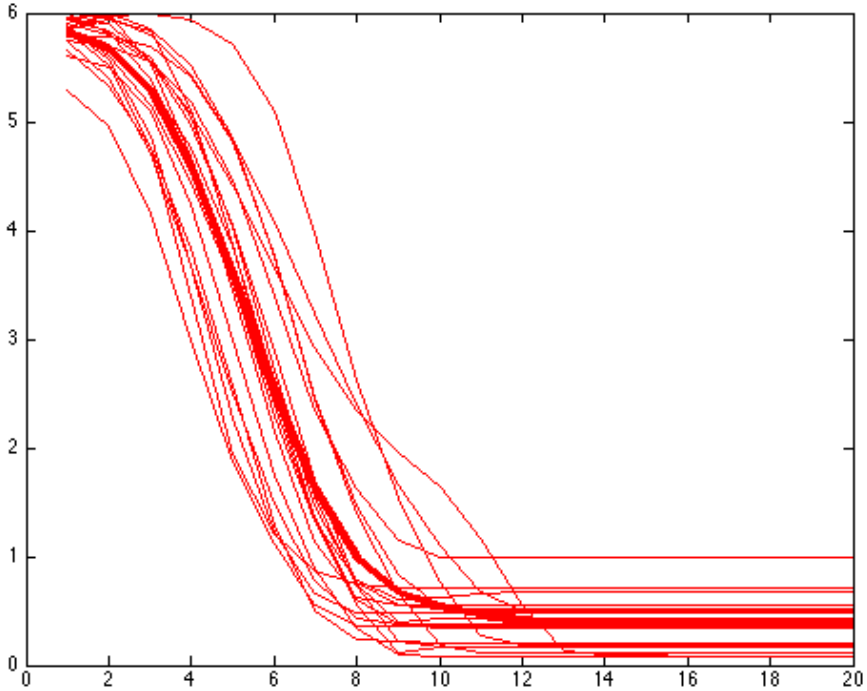
Expectation Maximization. (K=10, 2 true Gaussians)



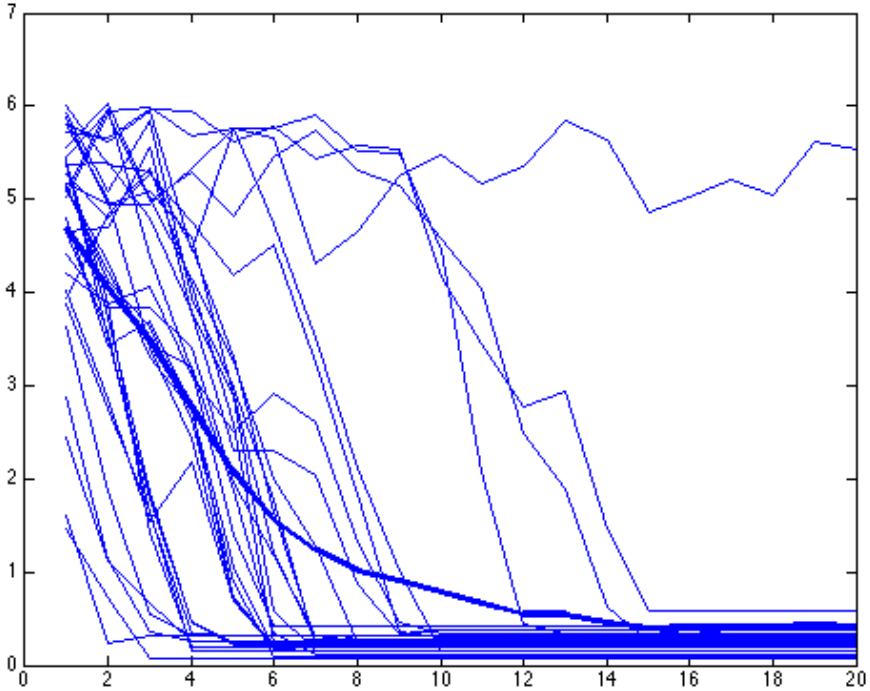
Variational Inference: Gaussian Mixture Model

Convergence speed vs. iterations

Variational Bayes



Gibbs Sampling



References

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